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PALEOMAGNETISM AND PALEOGEOGRAPHY OF THE NORTHERN CALCAREOUS ALPS (AUSTRIA)

J. E. T. Channell
Department of Geology, University of Florida, Gainesville

R. Brandner and A. Spieler
Institut für Geologie und Paläontologie, Universität Innsbruck, Innsbruck, Austria

J. S. Stoner
Department of Geology, University of Florida, Gainesville

Abstract. Prefolding magnetization components have been resolved at 38 sites of Liassic "Adnet" limestones in the central Northern Calcareous Alps (NCA) and at one site in the Drauzug. The site mean directions are northeasterly and down, and are closer to coeval European directions than to Southern Alpine "African" directions. The western end of both the NCA and of the Drauzug yield different site mean directions and negative fold tests indicating remagnetization. The sampling localities have been rotated relative to one another during thrusting, however, comparison of paleomagnetic poles with the European apparent polar wander path implies an average clockwise rotation of about 30° for the NCA relative to Europe. Larger-scale clockwise rotations in excess of 80°, relative to the Southern Alps, are implied by comparison with the Southern Alpine (African) polar wander path. Facies analysis of Jurassic and Lower Cretaceous basinal sediments in the central part of the NCA suggests that Liassic subsidence was related to presently E-W sinistral strike-slip faults, whereas Malm-Berriasian subsidence was controlled by presently E-W south-dipping normal faults. Both paleomagnetic and facies data tend to indicate that the NCA did not undergo the "African" rotation seen in Southern Alpine paleomagnetic data. This implies the existence of a short-lived tract of Triassic?-Jurassic ocean floor between the NCA and the Southern Alps, which linked the Pindos Ocean to the Ligurian trough and allowed Adria to rotate independently from the NCA during the Late Jurassic. Eo-Alpine deformation and metamorphism are associated with the Early Cretaceous closure of this "Hallstatt" trough.

INTRODUCTION

The first paleomagnetic study in the Northern Calcareous Alps (NCA) was that of Hargraves and Fischer [1959], who sampled Jurassic red Adnet Limestones and red radiolarites in the Scheibelberg close to Lofer (Figure 1). In this classic study, the authors supposed that hematite, of either detrital or early diagenetic origin, was the carrier of the remanent magnetization. They noted the similarity of the northeasterly natural remanent magnetization (NRM) directions to Jurassic poles for Europe available at that time. More recently, Mauritsch and Frisch [1978] sampled the Liassic Adnet Limestones west of Wolfgangsee (Figure 1) in the Osterhorn region. The mean direction (declination 62.5°, inclination 45.1°) was interpreted to indicate large-scale clockwise rotation of the NCA relative to Adria with the paleolatitude being compatible with the NCA as part of Adria prior to clockwise rotation. Similar declinations were also observed by Soffel [1979] from Skythian red sandstones in the central part of the NCA, although the mean inclinations are steeper than expected suggesting remagnetization. Northeasterly declinations were also obtained from seven sites in the Late Cretaceous and Paleocene Gosau Formation from the central NCA [Mauritsch and Becke, 1987], one of these sites has yielded a magnetostratigraphy across the Cretaceous/Tertiary boundary [Preisinger et al., 1986].

In contrast to the NE declinations observed in the central part of the NCA, Soffel [1975] reported northwesterly declinations in Upper Permian to Lower Triassic volcanic rocks from the Lechtal Alps (western NCA, Figure 1). Mauritsch and Becke [1987] also reported northwesterly declinations, from the western part of the NCA, recorded in Norian Plattenkalk and Hauptdolomite.

It has been almost universally accepted on facies grounds that the NCA were part of the Mesozoic continental margin of Adria, on the southern side of the Penninic Ocean [e.g. Tollmann, 1978; Frisch, 1979; Bosellini et al., 1980; Laubscher and Bernoulli, 1982; Weissert and Bernoulli, 1985; Lemoine and Trümpy, 1987; Tollmann, 1987a; Trümpy, 1988]. The northeasterly paleomagnetic declinations recorded in the central NCA [Mauritsch and Frisch, 1978] were interpreted to indicate large-scale (~90°) clockwise rotation of the NCA relative to Adria in Late Cretaceous time [Frisch, 1979]. The contrasting northwesterly declinations in the western NCA have been interpreted as due to straightening of an initially arcuate NCA about a pivot in the Inn Valley between Wörgl and Kufstein [Mauritsch and Becke, 1987; Flügel et al., 1987].

It has been recently suggested that paleomagnetic data and sedimentary facies are more compatible with a paleogeography in which the NCA were not part of Adria in Late Jurassic time [Channell et al., 1990]. We have expanded the paleomagnetic data set by including new data from the NCA and the Drauzug, with field tests to constrain the age of the magnetization components. We also present the results of new sedimentary facies analyses from the central NCA, and review the evidence for Mesozoic paleogeography of this unit.

GEOLOGICAL SETTING AND JURASSIC BASIN GEOMETRY

The Northern Calcareous Alps are composed mainly of Middle and Upper Triassic shallow-water limestones overlying Upper Permian and Scythian continental red beds. The Upper Triassic carbonates are sporadically overlain by a Jurassic/Cretaceous sequence of deeper-water limestones and siliceous sediments. The deeper-water sediments range in age from Hettangian (Liassic) to Berriasian (Lower Cretaceous) (Figure 2). The onset of basinal sedimentation is often marked by an abrupt break in the stratigraphy where light gray/white shallow-water Rhaetian limestones are overlain by Liassic red or dark gray basinal sediments. The Liassic sediments comprise dark-grey marly limestones (Allgäu Beds) deposited in basins, and red cephalopod-rich limestones (Adnet Limestone) deposited on swells and submarine highs. In the Callovian/Oxfordian, the subsidence reached its climax with the appearance of radiolarian cherts and siliceous limestones [Garrison and Fischer, 1969;

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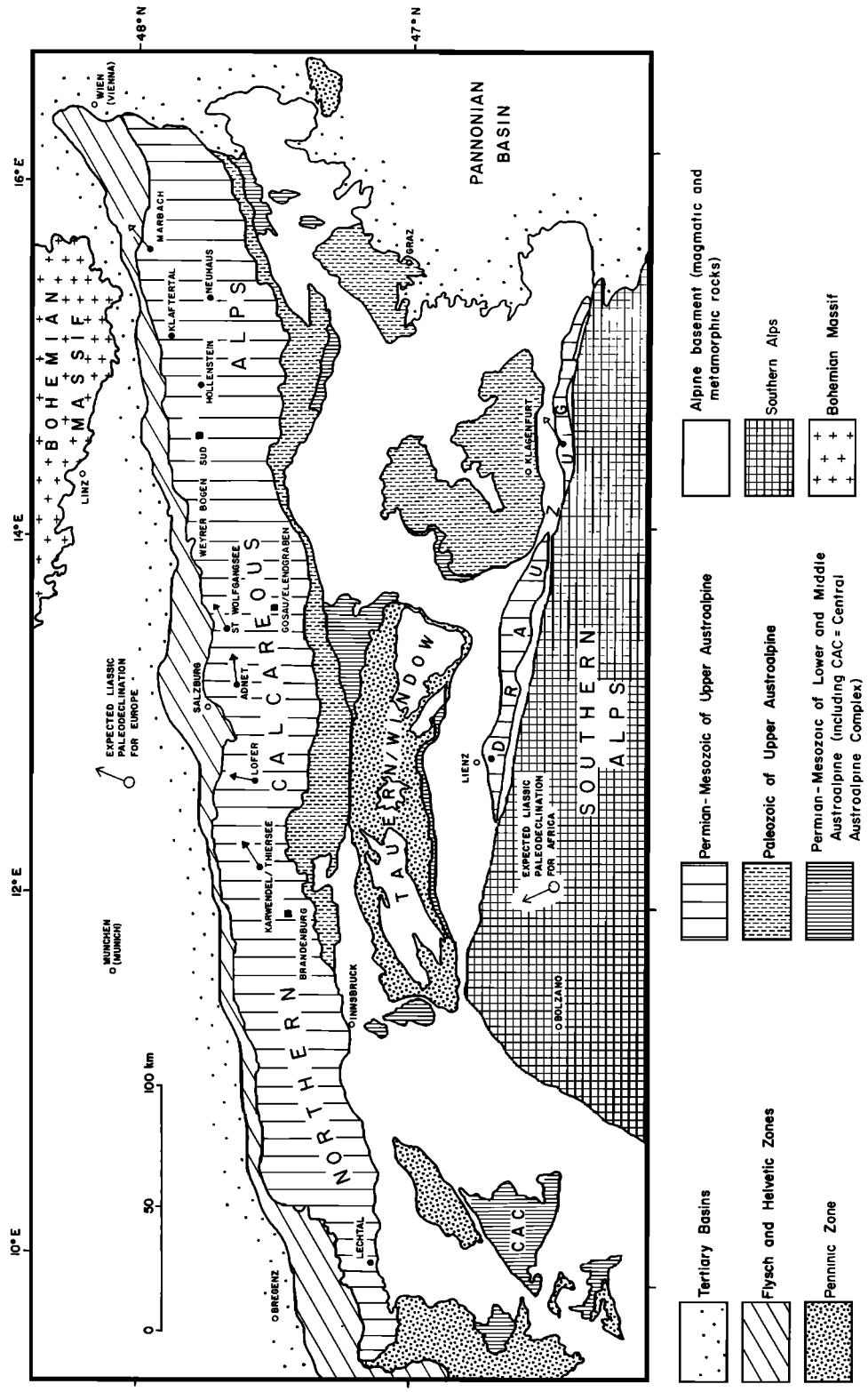


Fig. 1. Sketch map of the Northern Calcareous Alps. Solid circles indicate paleomagnetic sampling locations from this study. Mean paleodeclinations for sampling localities exhibiting "primary" magnetizations are indicated by arrows. Solid and open arrows denote means for groups of sites and individual sites, respectively. Solid squares indicate sampling localities in the Upper Cretaceous Gosau Formation of Mauritsch and Becke [1987].

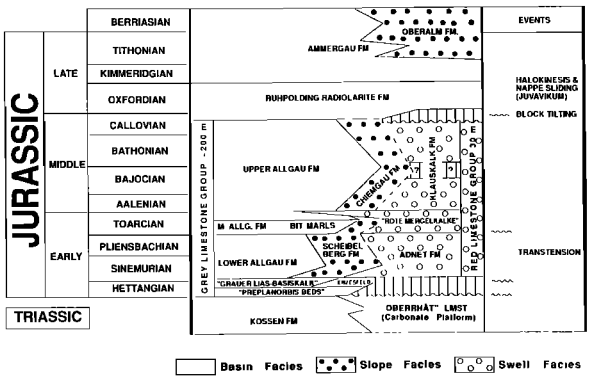


Fig. 2. Rhaetian to Berriasian stratigraphy in the Karwendel-Thiersee syncline.

Lein, 1987; Tollman, 1987b]. The upper part of the sequence comprises Berriasian marls of the Ammergau Formation (Figure 2).

Garrison and Fischer [1969] made the first modern facies analysis of Jurassic basinal sediments in the NCA. Utilizing the compensation depths for aragonite and calcite, they interpreted the facies in the Unken Syncline near Lofer (Figure 1) in terms of progressive Early Jurassic basin subsidence to abyssal depths by Oxfordian time, when radiolarites were deposited. In this area, basin slopes were recognized by large-scale slump folds and breccias with blocks up to tens of meters in diameter. Further to the east in the Glasenbach Gorge, Bernoulli and Jenkyns [1970] documented basin slope sedimentation in the Adnet Formation. Diersche [1980] established isopach maps of the Oxfordian radiolarites (Ruhpolding Formation) between Lofer and Wolfgangsee (Figure 1). The pattern was interpreted

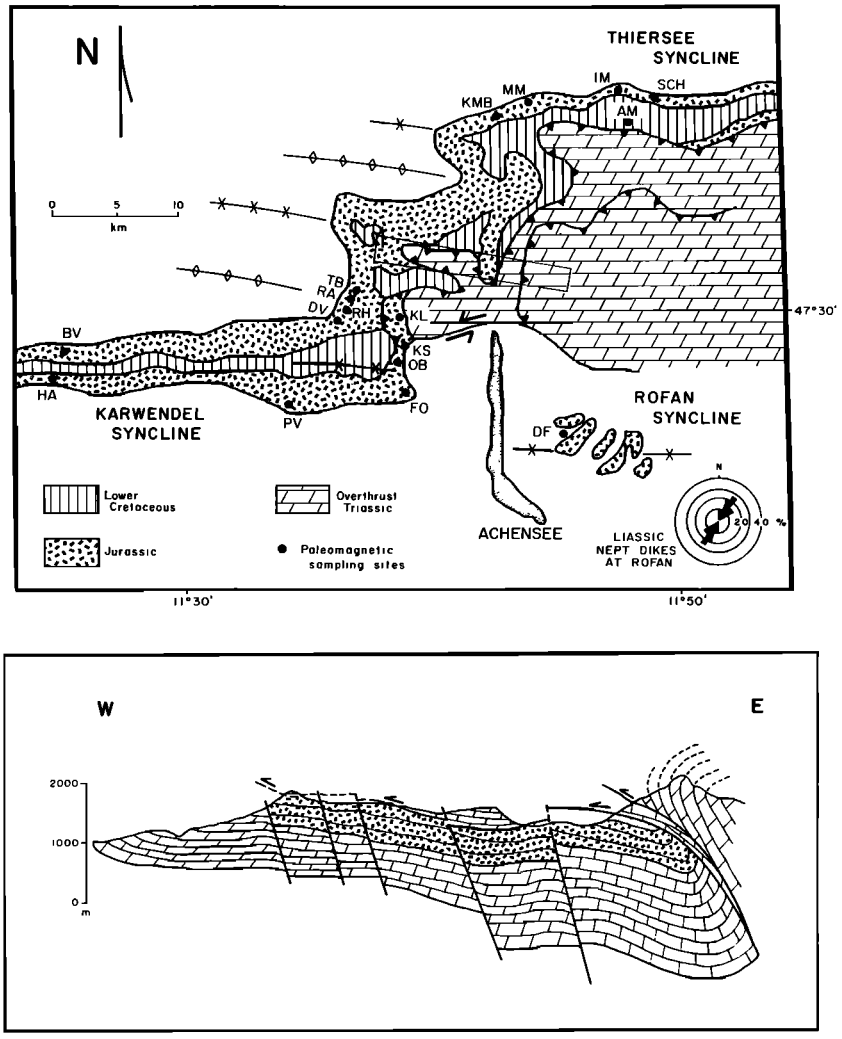


Fig. 3. Sketch map of the Karwendel-Thiersee syncline indicating the paleomagnetic sampling sites. For letter code to sites see Table 1. Orientation of major anticlines and synclines are shown by lines labelled with diamonds and crosses, respectively. The rose diagram indicates the orientation of neptunian dykes in the Rofan area, in the vicinity of site DF. The rectangular box in the central part of the syncline indicates the location of the cross section. Cross section oriented almost E-W, showing the Triassic shallow-water carbonates thrust westward over the Jurassic-Cretaceous basinal sediments.

in terms of presently E-W striking troughs and swells. In the same region, Vecsei et al. [1989] considered that the radiolarites were deposited below the aragonite compensation depth but just above the calcite compenstion depth, in contrast to the earlier interpretation of abyssal depths by Garrison and Fischer [1969].

Until now there has been no attempt to reconstruct the details of the Jurassic basin geometry, due to discontinuous outcrop and to complications related to Alpine deformation. The deformation history must be investigated together with facies analysis in order to elucidate the basin geometry. During the last 5 years, this type of investigation has been carried out in two regions where the Jurassic sediments are best exposed, in the Karwendel-Thiersee syncline and in the Unken-Diessbach region near Lofer (Figure 1).

The Karwendel-Thiersee syncline is now one of the best studied areas of Jurassic-Early Cretaceous basinal sediments. The syncline system is characterized by two E-W oriented synclines joined by a short N-S oriented segment to produce a S-shaped outcrop pattern (Figure 3). The central part of the syncline system is overthrust from the east by shallow-water Triassic sediments. Nagel [1975] documented the principal structural features of the region and noted the contrasting thicknesses of basinal sediments between the Thiersee syncline and the central part of the basin system. More recently, detailed facies analysis has shown that the central part of the syncline system is characterized by accelerated subsidence, particularly in Toarcian and Callovian-Oxfordian time [Spieler, 1992]. Maximum Liassic subsidence in the central part of the basin system is documented by upper Pliensbachian gravity flows followed by lower Toarcian black shales of the middle Allgäu Formation, which indicate poorly oxygenated localized deep basins (Figure 4). Coeval deposition of Bositra-rich red Adnet

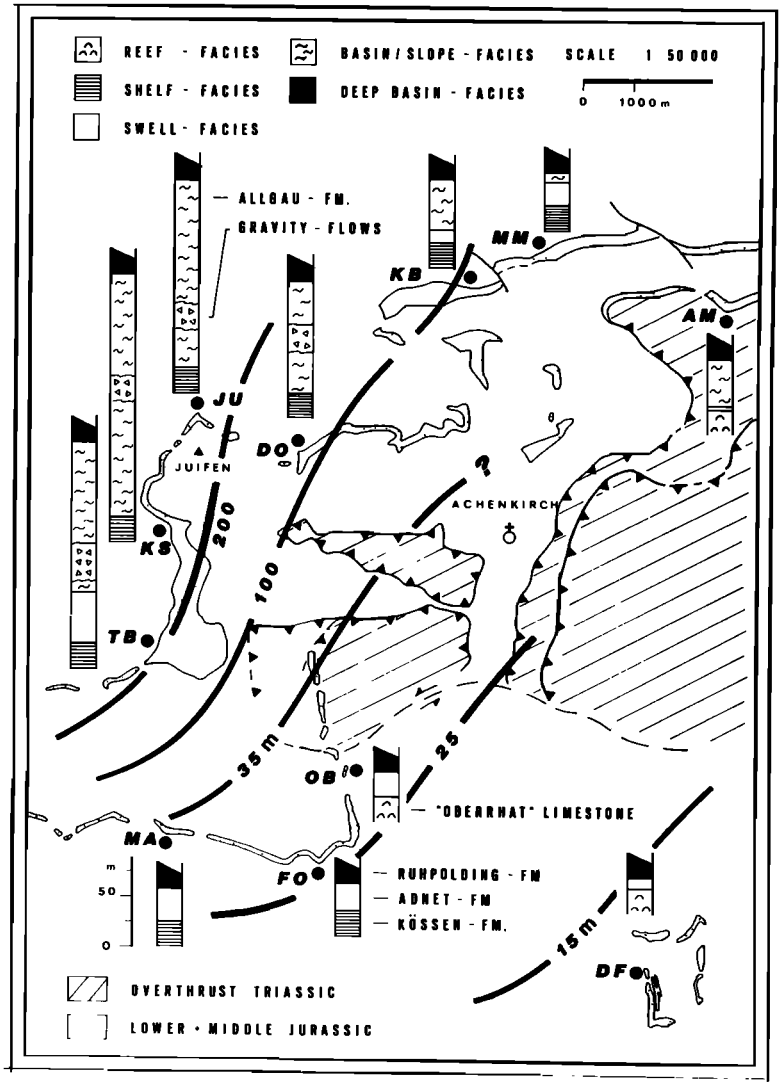


Fig. 4. Isopach map of Lower and Middle Jurassic sediments in the central part of the Karwendel-Thiersee syncline. Alpine tectonics are not restored. Abbreviations are: MA, Mantschenhalsl (thickness is 33 m); FO, Fonsjoch (27 m); OB, Oberautal (26 m); DF, Dalfazalm (11 m); AM, Ampelsbach (50 m); MM, Mahmoosbach (32 m); KB, Klammbach (62 m); DO, Dollmannsbach (110 m); JU, Juifen (185 m); KS, Kesselbach (240 m); and TB, Tiefenbach (200 m).

Limestones outside the central area of the syncline system indicates that accelerated subsidence was a local feature and not a characteristic of the entire basin. The Liassic facies distribution has been interpreted to indicate subsidence in a presently E-W oriented sinistral strike-slip fault system with the central part of the syncline system being a pull-apart basin [Channell et al., 1990; Spieler, 1992]. Above a very condensed Dogger section, the basin system is characterized by widespread distribution of Oxfordian radiolarites (Ruhpolding Formation) indicating a more regional acceleration in subsidence at this time. Liassic neptunian dykes, well exposed in the nearby Rofan region (Figure 3), which was a structural high (seamount) in Liassic and Dogger, are oriented NE-SW implying a NW-SE extension direction [Wächter, 1987]. This extension direction is consistent, in a transtensional kinematic framework, with the present E-W orientation of the proposed Karwendel-Thiersee sinistral strike-slip fault zone.

Liassic neptunian dykes are also well exposed in the Diessbach area (Figure 5) where they are oriented NNW, about 60° different from the more NE orientation of dykes of the same age in the Rofan area. Scarp breccias wedge out westward from N-S oriented normal faults which define the Unken and Diessbach basins (Figure 5). Onlap of the upper Liassic Allgäu Formation in the Diessbach basin indicates progressive filling of the basin at this time. In the Unken syncline, the breccias associated with the normal faults were deposited until Oxfordian, and at the eastern basin margin (Grubhörnl, Figure 5b) the onlap is Malm in age. In Oxfordian time, the basin geometry was significantly modified by extension accommodated by presently E-W oriented normal faults, one of which is recognized to the north of the Unken syncline (Figure 5b). Oxfordian to Tithonian movement on this fault is documented by thickening of sediments of this age towards the fault. Similar thickness variations are recognized further to the east between Adnet and Wolfgangsee (Figure 1) [Schlager and Schlager, 1973; Diersche, 1980; Plöchinger, 1990]. These presently E-W normal faults were often reactivated as thrust faults during Tertiary Alpine deformation. In the Unken syncline area, the E-W normal fault became the boundary between the Staufen-Höllengebirgs nappe and the Lechtal nappe. On the eastern side of the Grubhörnl, the N-S normal faults were reactivated during Eo-Alpine westward vergent compression, coeval with similar inversion in the Karwendel-Thiersee area (Figure 3).

The Oxfordian tilting on E-W oriented normal faults is widely recognized throughout the NCA by variations in sediment thicknesses and facies distribution. The tilting gave rise to basins which deepened to the north and were uplifted to the south, as indicated by subaerial erosional unconformities overlain by Malmian shallow-water carbonate platforms which shed shallow-water debris to the north. This debris is incorporated in the Oberalm Formation. The well-documented gravity sliding, in the central and eastern part of the NCA, of Hallstatt facies blocks into the Oxfordian radiolarite (Ruhpolding Formation) basins [Tollmann, 1981, 1987b; Mandl, 1982; Plöchinger, 1979, 1990] may be related to this large-scale block tilting. The debris in the Oberalm Formation includes Permian sandstone and evaporite detritus; leading to the conclusion that halokinesis is associated with the block tilting and gravity sliding.

The Jurassic stratigraphy of the Drauzug (Figure 1) indicates a basinal development very similar to that of the NCA. An evolution from Adnet to Allgäu and Ruhpolding formations, including thick scarp breccias, can be observed in basinal sequences exposed close to Lienz (Figure 1). The swell facies in this area are represented by Liassic Adnet Limestones and Dogger-Malm Biancone-like limestones. Sedimentation in the Liassic appears to be controlled by N-S normal faults [Blau and Schmidt, 1988] analagous to the the situation in the Unken-Diessbach region.

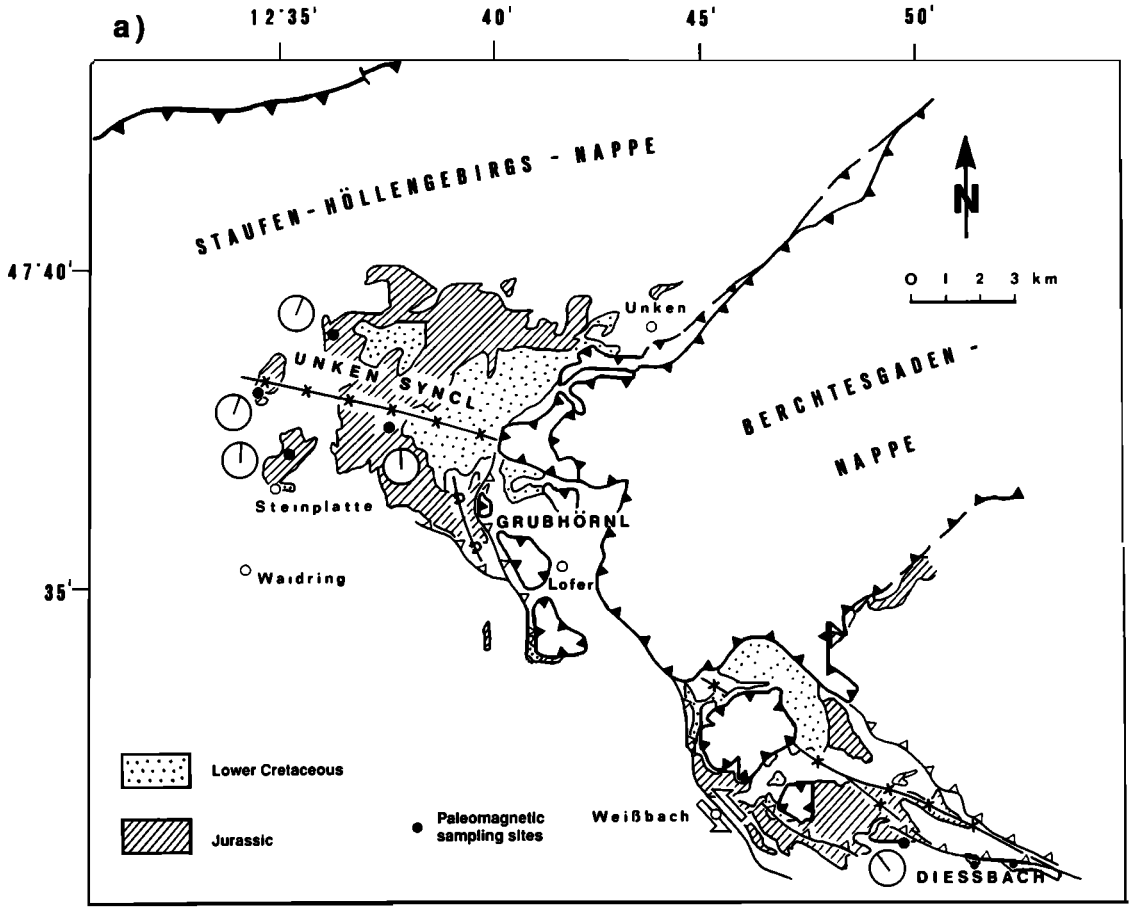
The Jurassic basin development in the NCA can be subdivided into two principal phases. The Liassic and Dogger subsidence is considered here to be controlled by presently E-W oriented sinistral strike-slip faults, with areas of accelerated subsidence (pull-apart basins) bounded by N-S oriented normal faults. The Malmian subsidence, on the other hand, appears to be controlled by presently E-W oriented normal faults, some of which may have had a Liassic strike-slip fault history. The Malmian normal faulting which is apparent from Oxfordian time is coeval with the onset of oceanic rifting in the Penninic-Ligurian-central Atlantic realm, and indicates a change in stress field at this time.

DEFORMATION

The internal nappe structure of the NCA was essentially formed by the important "pre-Gosau" tectonic phase which began in the Early Cretaceous. The onset of the compressional deformation is marked by Valanginian-Hauterivian synorogenic sedimentation of the Rossfeld Formation [Decker et al., 1987] and the cessation of deformation is stratigraphically constrained by the unconformably overlying Coniacian-Eocene Gosau Formation. The "pre-Gosau" deformation is characterized by presently-westward directed thrusting, as indicated by numerous shear criteria [Ratschbacher, 1987; Ring et al., 1989; Schmid and Haas, 1989; Ratschbacher et al., 1989; Platt et al., 1989]. The Gosau Formation comprises a heterogeneous sequence of deep-water shales, marls and intercalated turbidites, sandstones and conglomerates, with coral reefs, bioclastic limestones and coal deposits [Faupl et al., 1987]. The basal conglomerate of the Gosau Formation often transgressively overlies the nappe boundaries, forming a major unconformity [Tollmann, 1987a]. The Gosau basins have been associated with unroofing extension due to relaxation of the thickened imbricate thrust stack [Ratschbacher et al., 1989]. The Eocene ("post-Gosau") phase of deformation is marked by the termination of sedimentation in the Flysch Zone and the Helvetic Zone, as these units became involved in the southward-dipping subduction and progressively underthrust the NCA nappe pile. In the eastern Alps, this phase of collisional tectonics is characterized by presently N-S shortening, with eastward continental escape accommodated by sinistral strike-slip and normal faults becoming important by Miocene time [Ratschbacher et al., 1991].

Nagel [1975] documented the principal structural features in the Karwendel-Thiersee syncline and considered the central N-S oriented part to be a result of deformation of an originally rectilinear syncline, this deformation being coeval with the westward overthrusting of the shallow-water Triassic sequence. It was therefore considered that the E-W oriented folds predate the westward overthrusting of the shallow water Triassic. This interpretation is similar to that previously given by Ampferer [1941].

As discussed above, basinal sediments in the central part of the Karwendel-Thiersee syncline are considered to have accumulated in a Liassic pull-apart basin. The inversion of the pull-apart structure in the Early/Middle Cretaceous ("pre-Gosau") de-



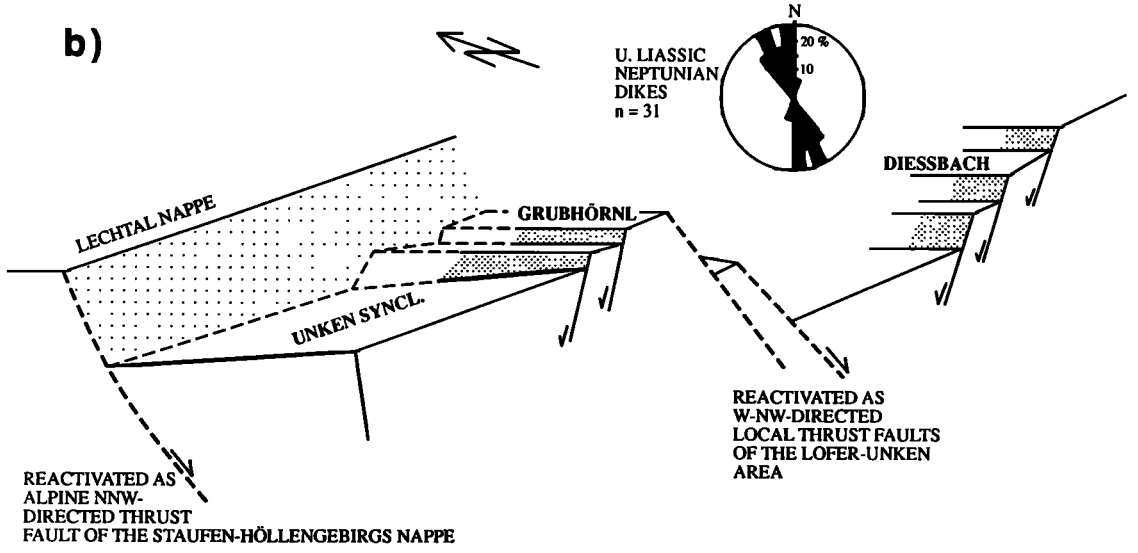


Fig. 5. (a) Sketch map of the Lofer (Steinplatte) and Diessbach sampling areas. (b) Interpretation, based on facies analysis, of the complex juxtaposition of NNW-SSE Liassic growth faults and WSW-ENE Oxfordian-Tithonian normal faults which were often reactivated by later thrusting. Rose diagram indicates the orientation of neptunian dykes at Diessbach.

formation phase resulted in the reactivation of presently E-W Jurassic sinistral strike-slip faults as dextral faults, resulting in westward thrusting of Triassic shallow-water carbonates over the Jurassic/Cretaceous basinal sediments (Figure 3). This direction of overthrusting is apparent from north-westward oriented fault striae [Eisbacher and Brandner, in prep.]. Recent mapping of the Karwendel-Thiersee syncline has shown that the westward overthrusting predates the E-W oriented folds which produced the synclinal structure of the.Karwendel-Thiersee region (Figure 3) [Spieler, 1992]. The westward thrusting and the E-W oriented folds can be associated with the "pre-Gosau" and "post-Gosau" deformation phases, respectively.

Tollmann [1987b] interpreted NW vergent recumbent folds involving Oxfordian radiolarites in the Rofan area (Figure 3) in terms of Oxfordian gravity sliding. However, it is now clear that these folds are overthrust (not depositionally overlain) by Malmian sediments and that the deformation is associated with post-Malmian ("pre-Gosau") thrusting.

PALEOMAGNETIC DATA

The paleomagnetic data discussed in this paper are derived from the Liassic Adnet Limestones. They are combined with and augment the data from 23 sites published by Channell et al. [1990]. In the combined data set, 38 sites are considered to carry primary magnetizations. For most of these sites, this conclusion is backed by fold tests and a conglomerate test. Five sites from the western NCA (Lechtal Alps) and the western Drauzug (near Lienz) do not yield primary magnetizations, as indicated by failed fold tests in these areas.

The majority of the sites are from four localities (Figure 1): Karwendel-Thiersee syncline (17 sites) (Figure 3), Lofer [Steinplatte] (eight sites) (Figure 5), Wolfgangsee (eight sites) and Adnet (three sites). Field tests support a primary magnetization component at all four localities. Additional sites were collected from Drauzug (three sites), Lechtal Alps (three sites), Marbach, Klaftertal, Hollenstein, Neuhaus and Diessbach (Figures 1 and 5).

Primary Magnetizations

Progressive thermal demagnetization of samples of Adnet limestone (Figures 6 and 7) indicates the presence of a well-defined high unblocking temperature magnetization component with maximum unblocking temperatures in excess of 600°C, indicating a hematite carrier. For most sites, this component has dual polarity. Alternating field demagnetization of typical Adnet limestone samples indicates that this magnetization component has high coercivity (Figure 8), consistent with a hematite carrier. The hematite is probably authigenic, growing during diagenesis and giving the limestones their reddish color. In view of the probable diagenetic origin of the remanence carrier, it is critical to constrain the age of magnetization components using field tests.

The high unblocking temperature magnetization component was computed by picking the relevant demagnetization temperature range from straight-line segments in the orthogonal projections, and using principle component analysis [Kirschvink, 1980] to compute the component direction. Component directions from individual sites were then combined using Fisher [1953] statistics (Tables 1 and 2).

For the Karwendel-Thiersee, Wolfgangsee, and Lofer (Steinplatte) areas, the site mean directions yield regional fold tests (Figures 9-11, Table 3) which are positive at the 99%, 99% and 90% confidence level [McElhinny, 1964], respectively. The high unblocking temperature magnetization components at these three localities therefore predate the latest (Eocene) deformation.

The paleomagnetic declinations in the Karwendel-Thiersee syncline are uniform (Figures 3 and 9, Table 1), apart from sites KL and KS which are located close to thrust faults and may therefore be affected by local rotation. The general uniformity in declination, particularly for sites located in the central part of the syncline system (namely sites DV, RA, RH, TB), suggests that the S-shape of the syncline system is a primary feature of the depositional basin, rather than a result of subsequent deformation.

Fold tests are not available for the type locality at Adnet, but the site offers a conglomerate test owing to the presence of the upper Pliensbachian "Scheck" breccia. This breccia is composed of clasts of Adnet Limestone, crinoidal Hierlatz Limestone, and pink crinoidal calcilutite and is cemented by white calcspar. The breccia is thought to be a result of turbidite flows on the edge of seamounts [Garrison and Fischer, 1969; Bernoulli and Jenkyns, 1974; Plöchinger, 1980; Wächter, 1987]. Thermal demagnetization of 14 clasts of Adnet Limestone from the breccia give resolvable high unblocking temperature magnetization components (Figure 12). The directions of the highest unblocking temperature component from the clasts are scattered (Figure 13), but are not random at the 95% probability level according to the Watson [1956] test for randomness. The orthogonal projections of the thermal demagnetization data (Figure 12) illustrate that the remanence in all clasts is dominated by the easterly/down direction (after tilt correction) which is the characteristic direction for the in situ Adnet limestone. Only at the high demagnetization temperatures do we see the randomized direction which predates the formation of the breccia. In some clasts, the magnetization is dominated by the component which postdates the incorporation of the clast in the breccia, and the component that predates the formation of the breccia is not apparent (Figure 12). We interpret these data to indicate that the growth of authigenic hematite in the clasts spans the time prior to and post incorporation of the clasts into the breccia. The magnetization was therefore acquired early in the diagenetic history of the sediment.

One of the sites from the Drauzug (Wildensteiner waterfall, near Klagenfurt) (Figure 1) yielded a positive fold test at the 99% confidence level [McElhinny, 1964] (Figure 14 and Table 3). One site from the eastern NCA (Marbach) (Figure 1) does not yield a fold test; however, the site exhibits dual polarity (Figure 15).

Secondary Magnetizations

The importance of field tests to constrain the age of the magnetization in the Adnet Limestones is illustrated by the observation of negative fold tests in the western part of the Drauzug (near Lienz, at Rötenbach 1) (Table 4 and Figure 16) and in the western part of the NCA (at Spullersee, Lechtal Alps) (Table 4 and Figure 17). The failed fold tests clearly indicate that the characteristic magnetization postdates the folding at these sites. Neighbouring sites have similar magnetization directions to those which fail the fold tests (Table 4), and therefore we conclude that these sites are also remagnetized. Whereas, most sampling sites in Adnet Limestone exhibited magnetization components with dual polarity, the remagnetized

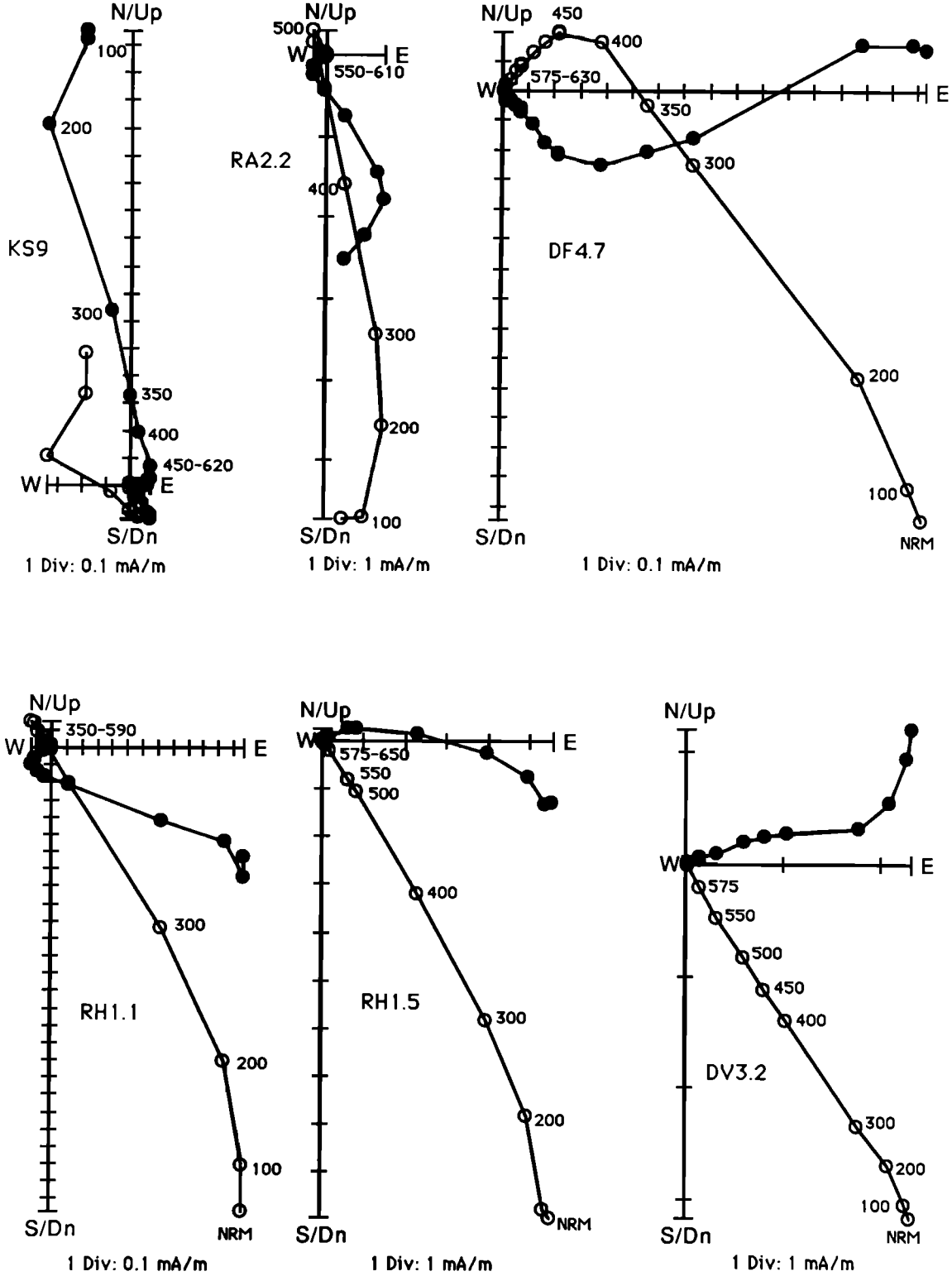


Fig. 6. Orthogonal projection of thermal demagnetization data for samples from the Karwendel-Thiersee syncline, after structural tilt correction. For letter code to sampling sites, see Table 1. Open and solid symbols represent projection on the vertical and horizontal planes, respectively. Specific demagnetization temperatures are given in degrees Celsius.

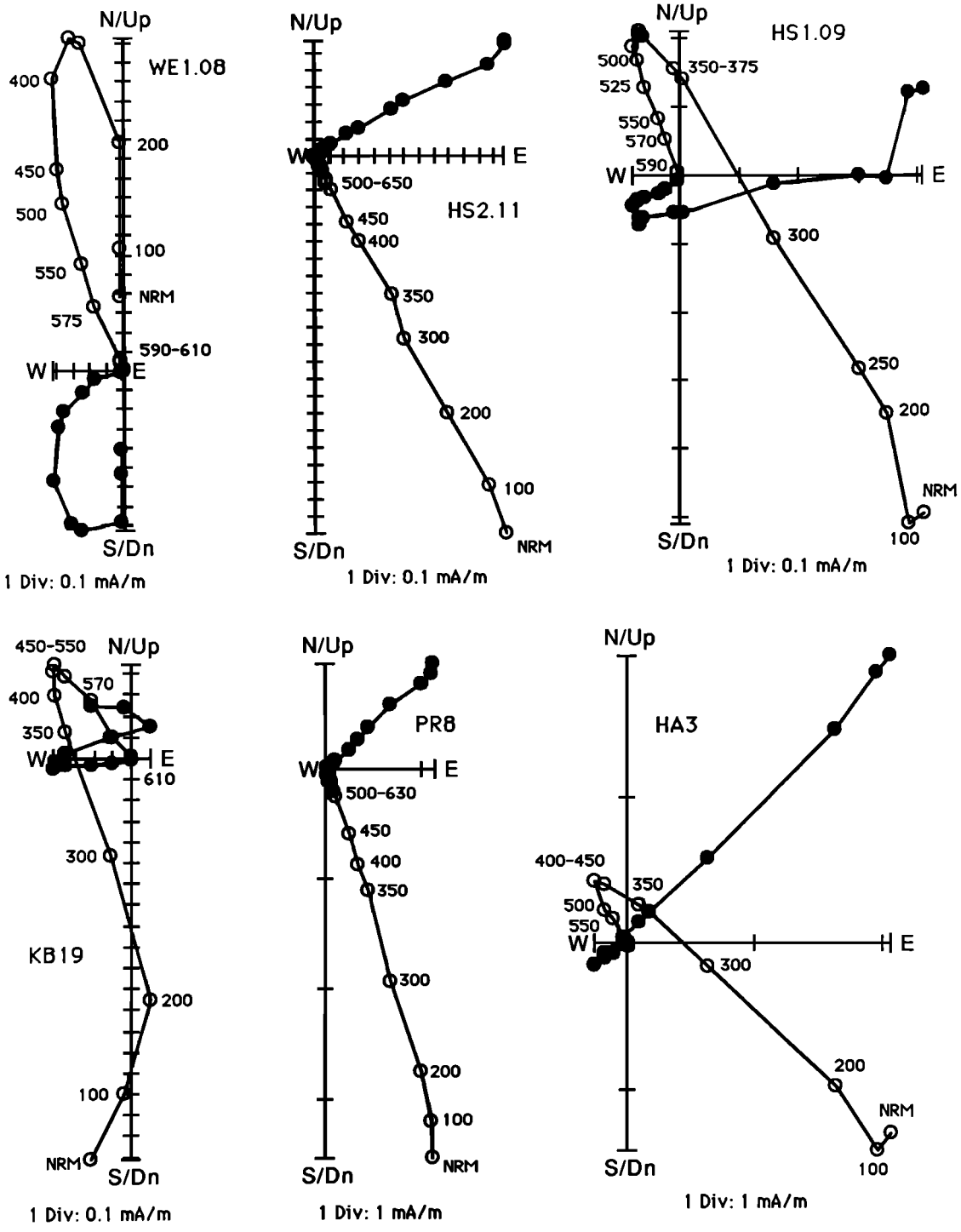


Fig. 7. Orthogonal projection of thermal demagnetization data for samples from the Wolfgangsee area, after structural tilt correction. For letter code to sampling sites, see Table 2. Notation is as for Figure 6.

sites all show single (normal) polarities. The thermal and alternating field demagnetization behavior at these sites is very similar to that seen in regions characterized by prefolding magnetizations (compare Figures 6, 7 and 18). We tentatively attribute this remagnetization to hematite growth associated with fluid migration/thermal events recognized from vitrinite reflectance and illite crystallinity data [Kralik et al., 1987; Krumm et al., 1988; Petschick, 1989]. Both indicators denote Late Jurassic - Early Cretaceous anchimetamorphism and epimetamorphism on the southern edge of the NCA, reaching a maximum in the Lechtal Alps. The low inclination of the postfolding magnetizations at Rötenbach (Lienz) and at Spullersee (Table 4) may

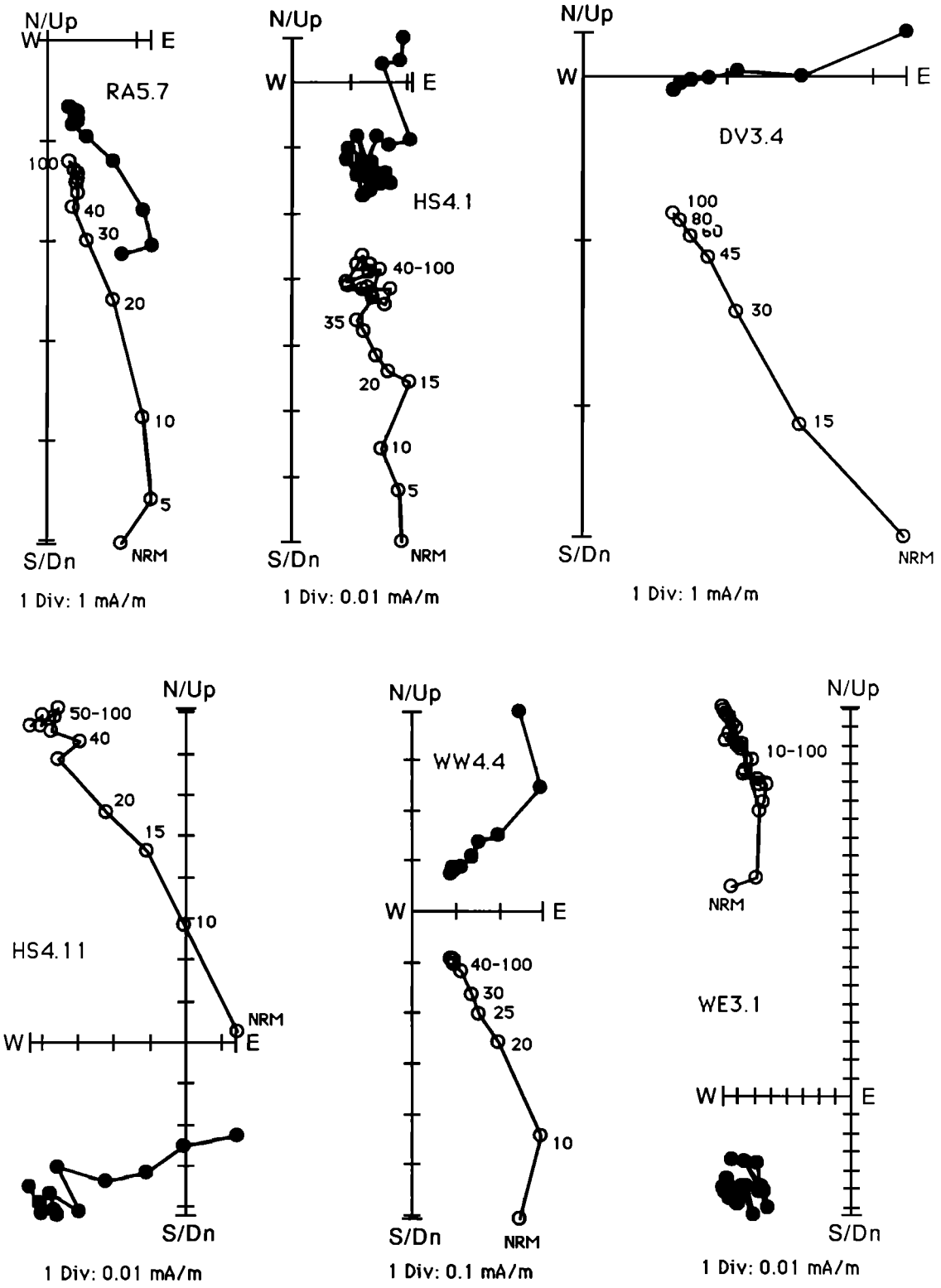


Fig. 8. Orthogonal projection of alternating field demagnetization for samples from the Karwendel-Thiersee syncline and the Wolfgangsee area after structural tilt correction. For letter code to sampling sites, see Tables 1 and 2. Open and solid symbols represent projection on the vertical and horizontal planes, respectively. Specific peak alternating fields are given in milliTesla.

TABLE 1. Site Mean Directions From the Karwendel-Thiersee Syncline

		Site Mean Before Tilt				Site Mean After Tilt			
Site	N/No	Dec. (°)	Inc. (°)	k	α95 (°)	Dec. (°)	Inc. (°)	k	α95 (°)
Kaserstatt (KS)	17/20	213.6	20.5	13.9	9.9	119.8	47.7	13.8	9.9
Retherhorn (RH)	13/15	222.3	-27.6	30.2	7.7	248.8	-47.3	30.1	7.7
Rethalm (RA)	14/19	40.1	25.5	30.3	7.3	71.5	58.3	30.3	7.3
Dürrach (DV)	18/18	299.9	-81.7	27.2	6.8	254.6	-55.9	27.2	6.8
Dalfazalm (DF)	17/23	191.8	-35.1	28.3	7.1	217.1	-59.4	30.8	6.8
Bayr Valley* (BV)	25/30	193.4	14.8	18.6	6.9	205.0	-62.2	18.6	6.9
Hoch Alpl* (HA)	26/27	161.7	10.2	6.0	12.7	30.4	58.6	7.0	11.6
Plumsbach* (PV)	13/15	175.4	15.4	14.0	11.5	63.0	25.2	14.0	11.5
Fonsjoch* (FO)	9/17	340.9	-29.1	5.9	23.2	278.5	-73.2	5.9	23.2
Oberautal* (OB)	20/37	244.2	19.8	9.3	11.3	238.9	-54.9	10.8	10.4
Kleekopf* (KL)	13/13	217.5	9.5	55.7	5.6	172.8	-54.9	49.6	5.9
Tiefenbachalm* (TB)	9/9	30.0	59.7	61.4	6.6	69.6	63.7	61.4	6.6
Ampelsbach* (AM)	10/14	300.6	41.4	7.6	18.7	271.5	-70.7	7.6	18.7
Klammbach* (KMB)	32/32	16.1	-1.0	42.6	3.9	39.5	60.6	41.5	4.0
Mahmoosbach* (MM)	19/19	16.5	6.6	43.1	5.2	44.1	48.9	43.1	5.2
Im Sattel* (IM)	11/11	7.3	14.1	71.6	5.4	41.0	51.6	71.6	5.4
Schönleiten* (SCH)	9/9	2.9	4.2	13.7	14.4	75.0	55.9	13.8	14.4
Locality mean	17	193.8	-1.7	3.9	20.8	59.8	58.5	18.4	8.5

Statistical parameters after Fisher [1953]. N/No indicates the ratio of samples collected to samples in which the high blocking temperature component was resolved. Dec is declination and Inc is inclination in degrees.

*Sites from Channell et al. [1990].

TABLE 2. Site Mean Directions From the Wolfgangsee Area

		Site Mean Before Tilt				Site Mean After Tilt			
Site	N/No	Dec. (°)	Inc. (°)	k	α95 (°)	Dec. (°)	Inc. (°)	k	α95 (°)
Wetzstein (WE1)	8/10	269.9	-48.6	211.3	3.8	252.7	-61.1	211.3	3.8
Wetzstein (WE2)	10/14	84.2	51.6	188.2	3.5	41.9	63.5	188.2	3.5
Wetzstein (WE3)	7/8	42.7	43.1	209.9	4.2	39.0	60.9	210.5	4.2
Kendlbach (KB)	19/19	60.4	53.9	47.0	4.9	68.1	65.0	47.1	4.9
Promeckbach (PR)	21/21	34.5	44.5	91.9	3.3	57.0	62.6	92.1	3.3
Hofwandalm (HA)	8/8	23.0	-71.0	11.3	18.8	257.4	-66.8	11.3	18.8
Hintersee (HS1)	12/13	81.9	61.5	57.6	5.8	71.6	56.4	57.6	5.8
Hintersee (HS2)	20/21	47.4	60.7	87.1	3.5	63.7	49.0	94.6	3.4
Locality mean	8	65.0	59.7	12.4	12.4	61.5	61.3	84.8	6.0

Codes are the same as for Table 1.

indicate remagnetization after the onset of deformation when the bedding was no longer horizontal. The declinations indicate variable amounts of postfolding rotation at different sites.

Three sites sampled in the eastern NCA (at Neuhaus, Hollenstein and Klaftertal) have different demagnetization behavior from the rest of the Adnet limestone sites. At these sites, a drop in magnetization intensity at about 300°-350°C and very erratic alternating field demagnetization data suggest iron sulfide remanence carriers. The resolvable component directions are generally very scattered, and we believe that the magnetization at these sites is secondary.

One site from Diessbach (Figure 5a) yields a NW mean magnetization direction (Table 4) that differs from the mean direction obtained from the nearby Steinplatte (Lofer) area (Figure 5a; Table 3). This may indicate local tectonic rotation at this site, however, we have no control on the age of magnetization. All magnetization directions have normal polarity at this site.

APPARENT POLAR WANDER PATHS

For those localities for which we have evidence for a "primary" magnetization (Table 3), we compare pole positions with coeval pole positions for Africa and Europe. The six Liassic pole positions from the NCA (Table 3) are significantly different from coeval poles on the synthetic European and African paths of Besse and Courtillot [1991] (Figure 19).

The Lofer (Steinplatte) region appears to have undergone anticlockwise rotation (of 14°) relative to Europe when compared with the European 200 Ma pole of Besse and Courtillot [1991], whereas the other sampling localities in the NCA appear to have undergone clockwise rotations of 31° (Karwendel-Thiersee), 33° (Wolfgangsee), 20° (Marbach), 52° (Adnet), and 27° (Wildensteiner, Drauzug) relative to Europe (Figure 19). A mean Late Cretaceous pole (at 57°N, 116°E; dp/dm=11°/16°) derived from the mean of four site means (Elendgraben, Gosau,

Brandenberg, and Weyrer Bogen Süd) (Figure 1) from Mauritsch and Becke [1987], implies a 38° clockwise rotation, when the pole is compared with the 70 Ma European pole of Besse and Courtillot [1991].

Comparison of the Liassic poles from the NCA with the 200 Ma African pole of Besse and Courtillot [1991] implies clockwise rotations relative to Africa of 36° (Lofer), 81° (Karwendel-Thiersee), 83° (Wolfgangsee), 69° (Marbach), 102° (Adnet), and 77° (Wildensteiner, Drauzug) (Figure 19). For the mean Late Cretaceous pole from the data of Mauritsch and Becke [1987], a fit to the 70 Ma (Late Cretaceous) African pole requires a clockwise rotation of 50°.

Although the Late Cretaceous paleomagnetic data of Mauritsch and Becke [1987] are not derived from the same thrust sheets as the Liassic data, the implication is that a large part of the observed clockwise rotations in the NCA are post-Late Cretaceous in age. It should be noted, however, that the now-obsolete blanket demagnetization techniques used on the Cretaceous sediments may not have isolated single magnetization components, and there are no field tests constraining the age of the magnetization.

The expected inclinations for NCA paleomagnetic poles to fit the Eurasian and African Liassic (200 Ma) poles are 59° and 49°, respectively. Mauritsch and Frisch [1978] noted that their mean inclination from the Adnet Limestones (45°) is more consistent with an African (Southern Alpine) origin for the NCA. In contrast, our mean inclinations from the Adnet Limestones (Table 3) are significantly steeper, and the overall mean for the four localities in the NCA (58.5°) is close to that expected for a European origin.

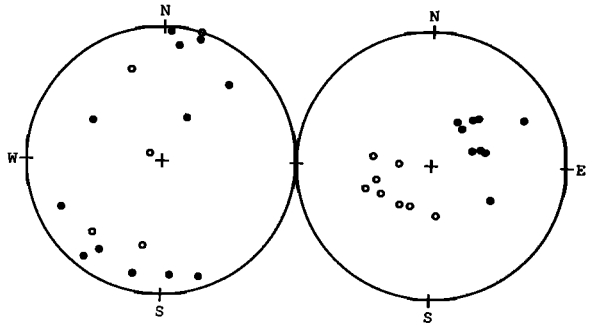


Fig. 9. Equal-area projection of site mean magnetization directions from the Karwendel-Thiersee syncline before (left) and after (right) structural tilt correction. Solid and open circles indicate downward (positive) and upward (negative) inclinations, respectively.

PALEOGEOGRAPHY

The Liassic and Late Cretaceous paleomagnetic data from the NCA are more compatible with coeval data from Europe, than with coeval data from Africa. In contrast, the Jurassic and Cretaceous paleomagnetic data from the Southern Alps are consistent with recently derived African apparent polar wander paths (see reviews by Channell et al. [1979], VandenBerg and Zijderveld [1982], Lowrie [1986], Channell [1986], Heller et al. [1989], and Channell et al. [1992]).

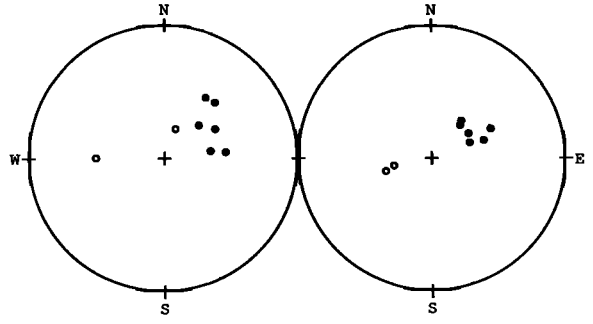


Fig. 10. Equal-area projection of site mean magnetization directions from the Wolfgangsee area before (left) and after (right) structural tilt correction. Notation is as for Figure 9.

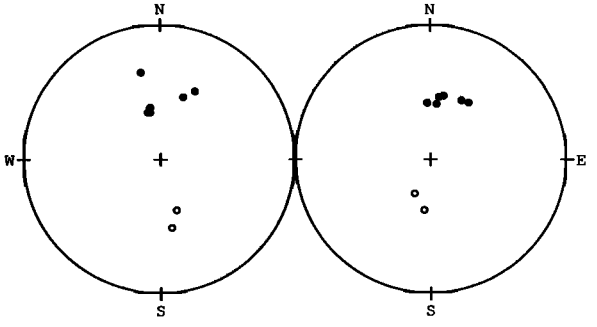


Fig. 11. Equal area projection of site mean magnetization directions from the Lofer (Steinplatte) region before (left) and after (right) structural tilt correction. Notation is as for Figure 9.

TABLE 3. Locality Mean Directions Interpreted as "Primary" Liassic Magnetizations

		Mean Before Tilt				Mean After Tilt				Pole Position			
Locality	Sites (Samples)	Dec. (°)	Inc.(°)	k	α95 (°)	Dec. (°)	Inc. (°)	k	α95 (°)	Lat °N	Long °E	dp (°)	dm (°)
Karwendel/Thiersee	17	193.8	-1.7	3.9	20.4	59.8	58.5	18.4	8.5	46.8	89.5	9.3	12.6
Wolfgangsee	8	65.0	59.7	12.4	16.4	61.5	61.3	84.8	6.0	48.1	81.6	7.6	8.1
Lofer *	8	357.6	52.4	31.4	10.0	15.1	55.0	71.2	6.6	73.3	144.2	6.7	9.3
Adnet *	3	79.4	47.5	149.2	10.1	80.8	57.1	63.9	15.5	32.3	80.6	16.5	22.5
Marbach	1 (11)	48.5	29.6	37.8	8.5	47.8	60.2	32.4	8.1	55.5	98.5	9.3	12.3
Wildensteiner	1 (23)	67.3	29.6	21.5	6.7	56.4	51.6	47.2	4.5	45.1	102.0	4.2	6.1

Statistical parameters are after Fisher [1953]. Lat is latitude and Long is longitude. Other codes are as for Table 1.

* Locality means from Channell et al. [1990]

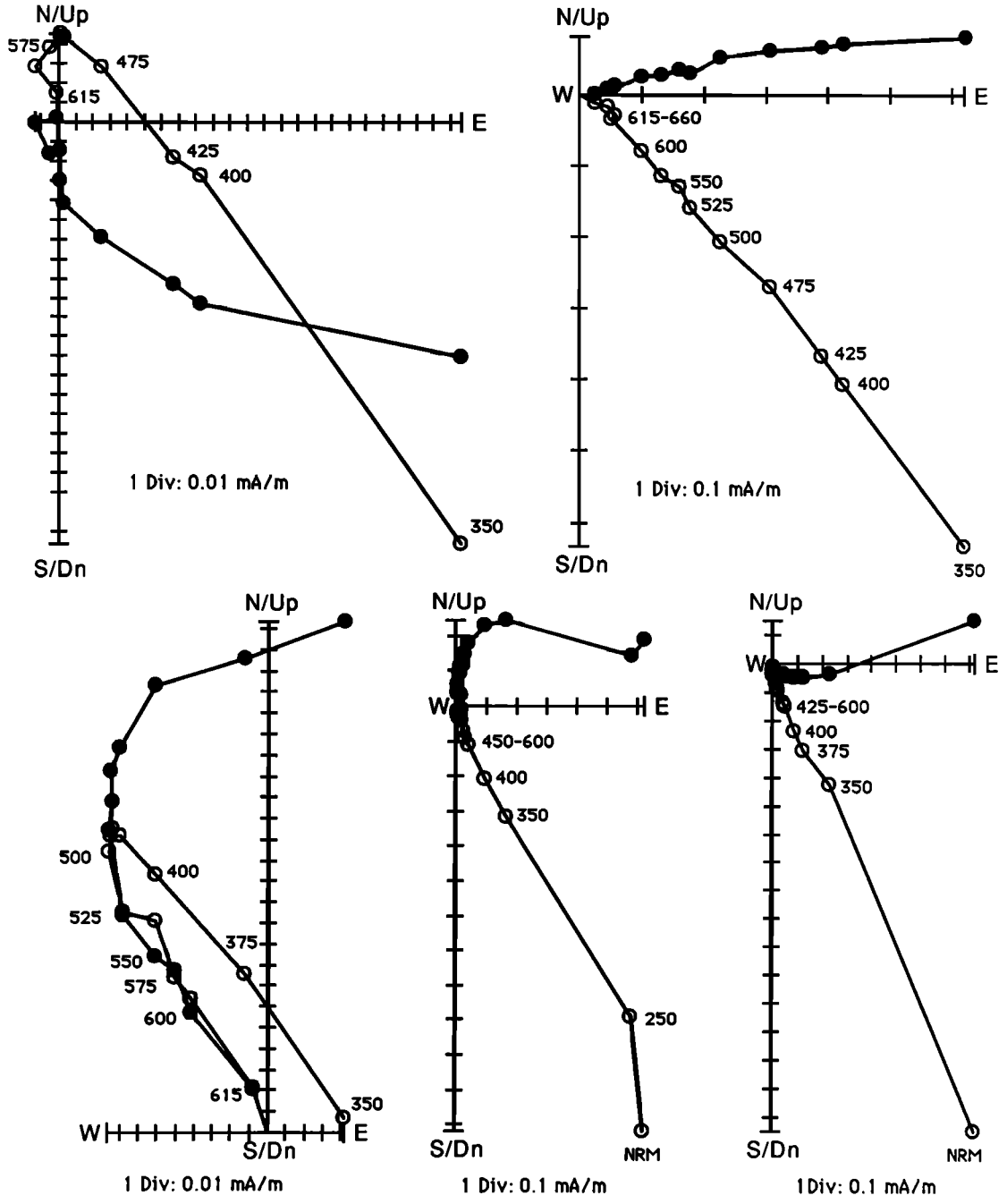


Fig. 12. Orthogonal projection of thermal demagnetization data of samples from the "Scheck" breccia at Adnet village after structural tilt correction. Notation is as for Figure 6.

On the basis of preliminary paleomagnetic and facies data, it was proposed that the NCA did not undergo the "African" rotational history observed in the Southern Alps but were separated from the Southern Alps by a branch of Jurassic Tethys (Channell et al., 1990). This short-lived ocean, referred to as the "Hallstatt" trough, was considered to have opened in Triassic? or Early Jurassic time, and closed during Early Cretaceous. The Penninic ocean between Europe and the NCA began to open in mid-Jurassic and was not fully closed until Eocene.

Several lines of evidence point to the existence of an early Mesozoic zone of mobility, partly floored by oceanic crust, at the present-day southern margin of the NCA. Eclogites within the southern Austroalpine crystalline basement (Koralpe, Kreuzeck, Texelgruppe) mark a zone of high-pressure amphibolite facies rocks with Nd and Sr isotopic ratios lying in the MORB field [Miller, 1990; Thöni, 1990]. These eclogites yield mineral cooling ages spanning the Late Cretaceous (see Miller, [1990]), consistent with Eo-Alpine metamorphism. However, in the southern Koralpe where the gabbro to eclogite transition

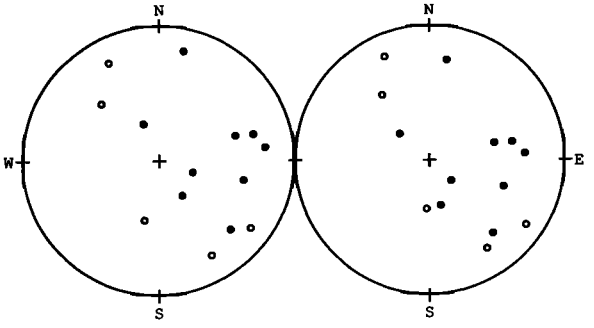


Fig. 13. Equal-area projection of sample magnetization directions from Adnet Limestone clasts in the "Scheck" breccia at Adnet village before (left) and after (right) structural tilt correction. Notation is as for Figure 9.

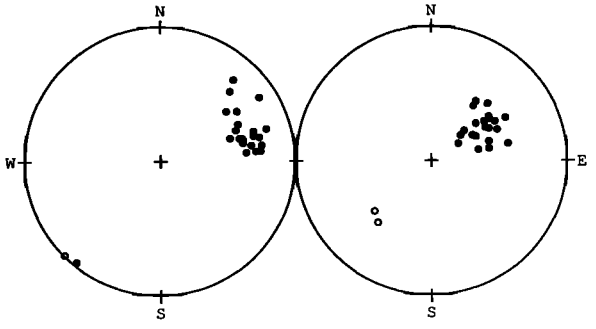


Fig. 14. Equal-area projection of sample magnetization directions from the Wildensteiner waterfall (near Klagenfurt, Drauzug) before (left) and after (right) structural tilt correction. Notation is as for Figure 9.

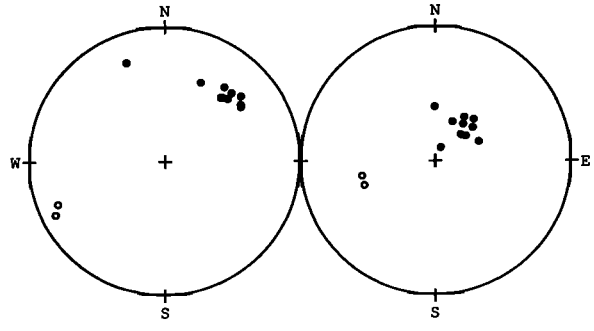


Fig. 15. Equal-area projection of sample magnetization directions from Marbach before (left) and after (right) structural tilt correction. Notation is as for Figure 9.

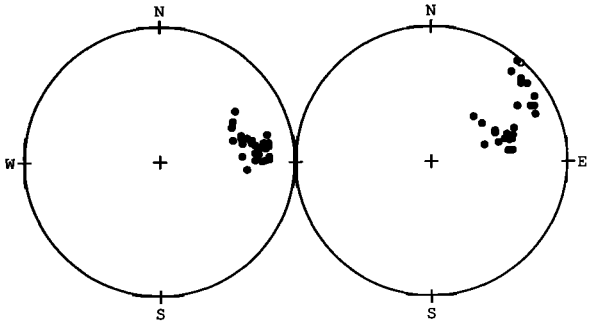


Fig. 16. Equal-area projection of sample magnetization directions from Rötenbach 1 (Lienz, Drauzug) before (left) and after (right) structural tilt correction. Notation is as for Figure 9.

is exposed, the gabbro has yielded a Sm/Nd mineral isochron of 275 Ma (+/- 18 my) [Thöni, 1990] raising the possibility that these eclogites were part of an ocean of Permian age. On the basis of eclogite occurrence and high pressure - high temperature metamorphism, the possible suture can be traced from the southern Koralpe in the east, through the Kreuzeck to the Texelgruppe (southern Ötztal nappe) [Thöni, 1980; Frank et al., 1987] and finally to the Sesia-Lanzo zone in the west. The high pressure assemblages in the Sesia-Lanzo zone are perhaps the best studied, as it is here that the Tertiary metamorphic overprints are less pronounced. Both basic and granitoid rocks have been transformed to eclogites in the Sesia-Lanzo zone, and the pressure peak is dated at about 110 Ma [Dal Piaz et al., 1972; Compagnoni et al., 1977; Hunziker et al., 1989]. The granitoid rocks affected by high pressure metamorphism in the Sesia-Lanzo zone are generally considered to have constituted Austroalpine basement. The subduction of continental basement to depths of 50-100 km is characteristic of continent-continent collision zones, but "Eo-Alpine" orogeny in the western and central Alps is often considered to represent the onset (not the culmination) of subduction of the Penninic ocean. This dilemma has been addressed by proposing bivergent subduction [Laubscher, 1970], two southward dipping subduction zones [Dal Piaz et al., 1972], subduction of continental crust during early stages of B-subduction [Burchfiel, 1980; Hsü, 1991], and by placing the Eo-Alpine subduction zone within the Austroalpine continental crust [Williams and Compagnoni,

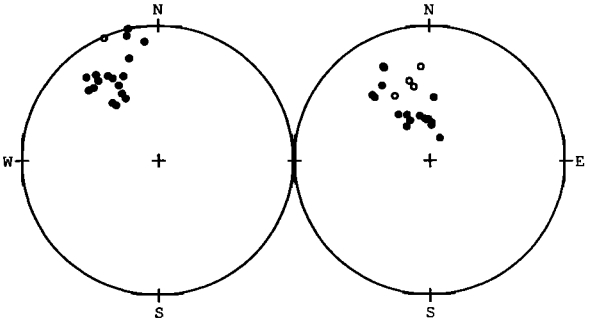


Fig. 17. Equal-area projection of sample magnetization directions from Spullersee (Lechtal Alps) before (left) and after (right) structural tilt correction. Notation is as for Figure 9.

1983; Rubie, 1984; Frank, 1987]. Alternatively, the underthrusting of Austroalpine continental basement may be associated with the culmination (continent-continent collision phase) of subduction of an ocean basin located to the present-day south of the Austroalpine units (Figure 20), the high pressure metamorphic peak at 110 Ma being consistent with Early Cretaceous closure.

Detrital chromium spinels in Cretaceous flysch deposits in the eastern Alps indicate eroding ophiolitic bodies to the

TABLE 4. Site Mean Directions Interpreted as Remagnetizations

Site	N/No	Site Mean Before Tilt				Site Mean After Tilt			
		Dec. (°)	Inc. (°)	k	α95 (°)	Dec. (°)	Inc. (°)	k	α95 (°)
Spullersee (SP)	18/18	332.5	27.5	20.4	7.8	339.1	36.5	3.9	20.3
Trittalm (TR)	14/14	279.0	43.2	10.1	13.1	281.4	60.0	10.2	13.1
Lorüns (LO)	21/23	190.8	53.3	179.3	2.4	304.1	58.5	179.3	2.4
Rötenbach (RO1)	27/27	80.0	29.8	57.7	3.7	61.9	31.5	16.0	7.2
Rötenbach (RO2)	22/22	65.2	38.3	36.0	5.4	55.0	33.1	37.2	5.2
Diessbach (DB)	19/19	302.9	63.9	57.1	4.5	326.4	43.5	57.2	4.5

Codes are the same as for Table 1.

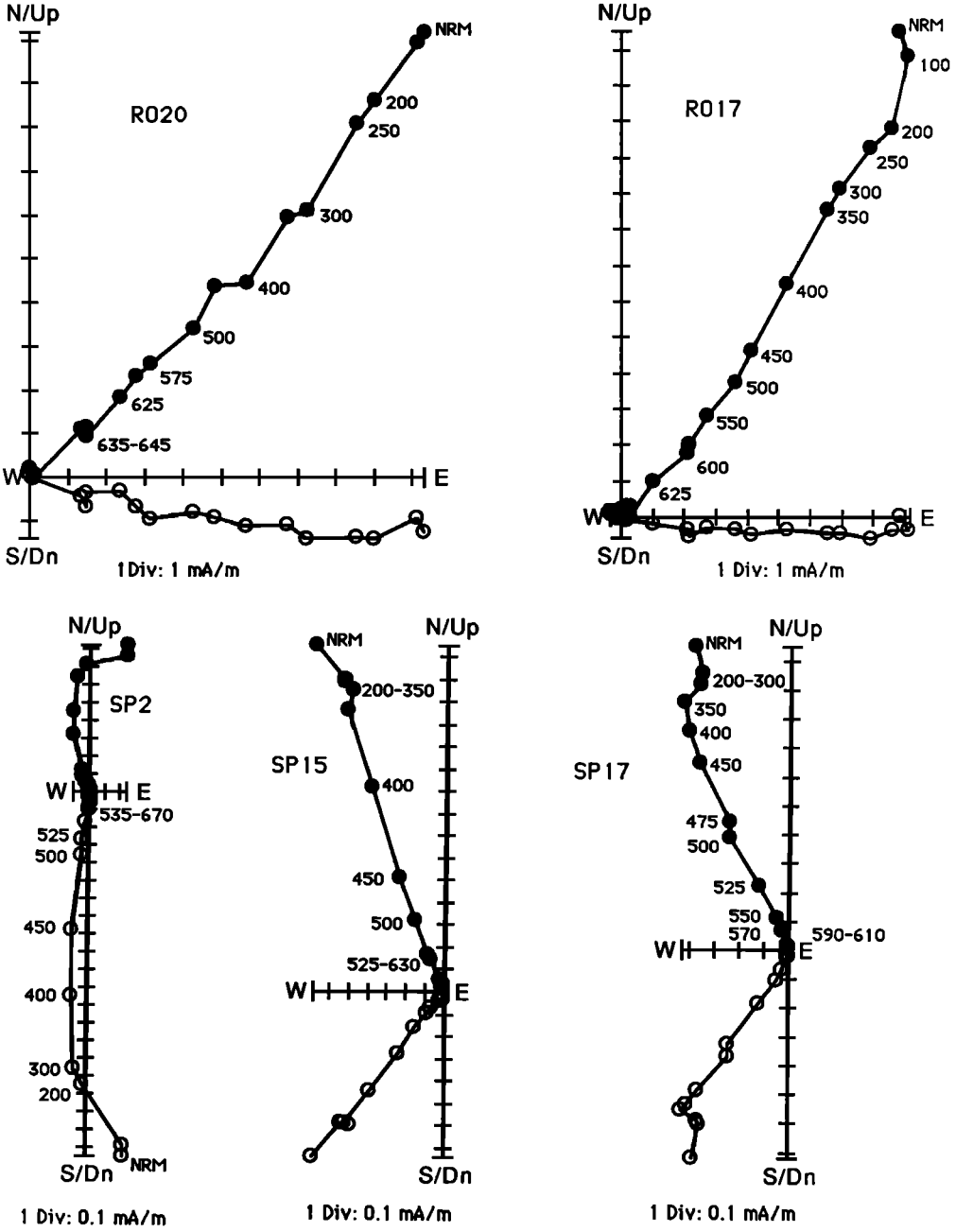


Fig. 18. Orthogonal projection of thermal demagnetization data of samples from Rötenbach (RO) (Drauzug, near Lienz) and Spullersee (SP) (Lechtal Alps) after structural tilt correction. Notation is as for Figure 6.

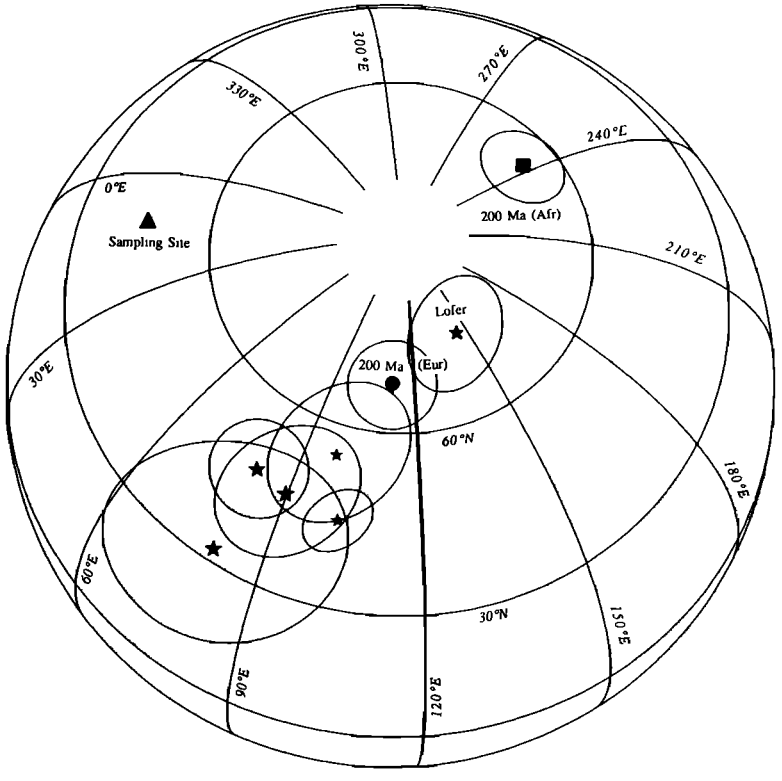


Fig. 19. Liassic pole positions from the Northern Calcareous Alps (stars) (Table 3) compared with the coeval (200 Ma) pole from the African (solid square) and European (solid circle) polar wander paths of Besse and Courtillot [1991]. Large and small stars denote multiple and single sites, respectively. 95% confidence limits are shown for each pole.

present-day north and to the present-day south of the NCA [Decker et al., 1987; Pober and Faupl, 1988]. Southern sources dominate in the Hauterivian-Albian Rossfeld and Lavant formations, and are apparent in parts of the lower Gosau and upper Gosau complexes. Northern ophiolitic sources are observed in the Albian-Cenomanian Losenstein Formation and in the Gosau complexes. Winkler [1988] reported detrital chromite in the Lechtal and Allgäu nappes and in the Arosa and Walsertal zones from Barremian/Aptian time and, perhaps more importantly, detrital glaucophane in the early Albian to Coniacian interval. As pointed out by Winkler [1988], the presence of high-pressure metamorphic minerals as detrital grains in mid-Cretaceous flysch points to a considerably older subduction phase. More localized occurrences of detrital chromium spinels are known in the Late Jurassic in the middle Penninic (Idalp sandstone) and lower Austroapline (Hippold nappe). For these deposits the ultrabasic host rocks may have ascended along Penninic transform faults and may have been reworked soon after their formation [Pober and Faupl, 1988]. This cannot be the case for the very widespread spinel detritus seen in the Early Cretaceous deposits. Earliest Cretaceous anchimetamorphism is apparent throughout much of the NCA from illite crystallinity data, with higher grade epimetamorphism associated with the southern rim of the NCA [Kralik et al., 1987]. The first hallmark of closure of the ocean to the present-day south of the NCA is manifest by the occurrence of detrital chromium spinels in the Valanginian-Hauterivian Rossfeld Formation. The Late Jurassic northward sliding and thrusting of the Hallstatt and Dachstein nappes is explained by strong block tilting and halokinesis in the middle part of the NCA. There are no indications of Late Jurassic compressional tectonics affecting the NCA.

Rather than a paleogeography in which the NCA were separated from the Southern Alps, the alternative, that the NCA were part of the same continental margin as the Southern Alps in Late Jurassic time, implies large-scale clockwise rotation (70°-100°) of the NCA during deformation. This would mean that the "Hallstatt" trough, represented by the eclogitic suture to the present-day south of the NCA, lay to the east of the NCA in Late Jurassic time, and may therefore have been a continuation of the Pindos/Vardar ocean tract. As discussed above, Late Jurassic facies distribution in the NCA indicates subsidence on presently E-W oriented, south dipping, normal faults; and the rapid regional subsidence is consistent with passive margin development until post-Berriasian time. Latest Jurassic onset of compressional tectonics on the margins of the Vardar ocean in the Dinarides and Hellenides (see Dercourt et al., [1986]) is apparently inconsistent with the "Hallstatt" trough being an rectilinear extension of the Pindos/Vardar ocean. We favor a paleogeography in which the region of extension to the present-day south of the NCA lay between the Southern Alps and the NCA and provided a Late Jurassic connection between the Ligurian and Vardar/Pindos oceans.

CONCLUSIONS

Jurassic sedimentation in the NCA can be divided into two principal phases. The Liassic-Dogger phase was controlled by

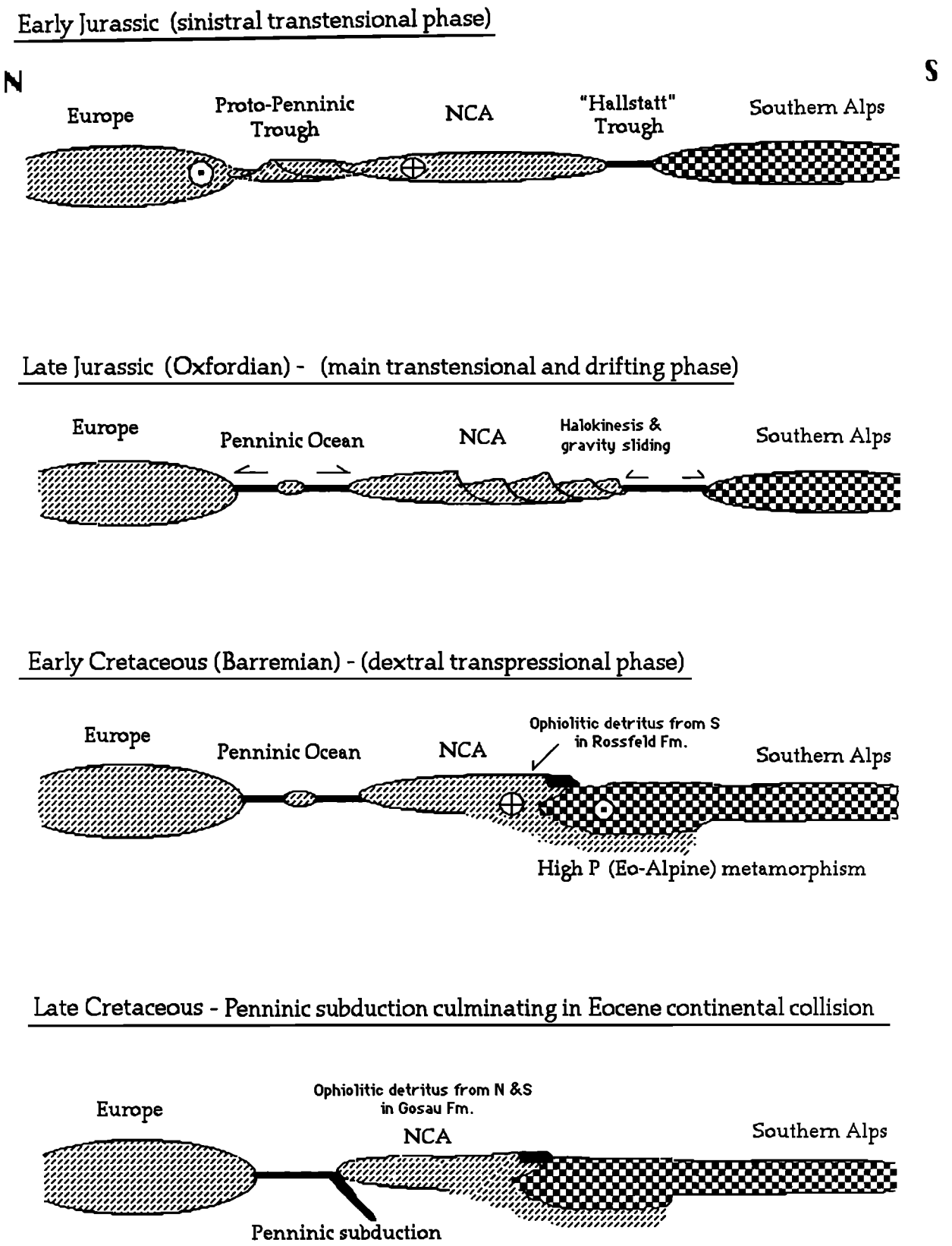


Fig. 20. Cartoon indicating the proposed evolution of a N-S section across the Eastern Alps. Bull's-eyes and crosses represent relative motion out of and into the plane of the cartoon, respectively.

presently E-W oriented sinistral strike-slip faults. The Malm phase was controlled by presently E-W oriented, south dipping, normal faults. The change in stress field is correlated to the onset of the oceanic phase of rifting in the Atlantic/Ligurian realm. The Oxfordian normal faults in the NCA may be linked to a southward dipping detachment fault in a simple shear rifting model [Wernicke, 1985; Froitzheim and Eberli, 1990]. This would imply that the southern NCA represented the lower plate margin to the "Hallstatt" trough during Oxfordian time.

Paleomagnetic data from the NCA are consistent with either large-scale (70°-100°) or small-scale clockwise rotation (20°-50°) of the NCA nappes, depending on whether the NCA were part of, or independent from, the Southern Alpine continental margin prior to deformation. If the NCA were separated from the Southern Alps during Jurassic time and did not undergo the "African" Mesozoic rotation observed in the Southern Alps, the small-scale rotations apply. The large-scale rotations would place the southern rim of the NCA to the east in Late Jurassic time, approximately colinear with the Vardar trough. As Late Jurassic sedimentation in the NCA is not compatible with latest Jurassic compressive deformation in the Vardar trough, we prefer a paleogeography for the NCA in which this unit was separated from the Southern Alps by a narrow "Hallstatt" ocean of Triassic? and Jurassic age, which linked the Ligurian ocean in the west to the Pindos/Vardar ocean in the east (Figure 20). The Penninic ocean developed in mid-Jurassic time between the NCA and Europe, isolating the NCA as an intermediate unit in the Jurassic Tethys. The closure of the "Hallstatt" trough in Early Cretaceous time culminated in ("pre-Gosau") Eo-Alpine continent-continent collision in which continental crustal rocks underwent high-pressure metamorphism.

The proposed paleogeography in which the NCA were part of an independent microplate in Late Jurassic time, and were not part of the Southern Alpine continental margin, may resolve an apparent dilemma. On the one hand, Permian, Jurassic and Cretaceous paleomagnetic data from the Southern Alps are consistent with recently derived African apparent polar wander paths implying motion of Southern Alps with Africa (see reviews by Channell et al. [1979], VandenBerg and Zijderveld [1982], Lowrie [1986], Channell [1986], Heller et al. [1989], and Channell et al. [1992]). On the other hand, Early Cretaceous Eo-Alpine deformation, characterized by westward-directed tectonic transport, is not compatible with the relative motion of Africa and Europe derived from Atlantic spreading history, implying that the Southern Alps were not part of Africa at this time [Platt et al., 1989; Hsü, 1989]. In the proposed paleogeography, Eo-Alpine deformation is associated with closure of an oceanic "Hallstatt" trough between the "NCA" microplate and the Southern Alps (Figure 20) and is therefore not a direct consequence of the relative motion of Europe and Africa. The Valanginian onset of rotation of the Iberian peninsula [Srivastava and Tapscott, 1986] may have modified the relative motion of continental microplates in the Penninic realm (see Coward and Dietrich, [1989]), possibly resulting in inversion of the relative motion of the "NCA" microplate and the Southern Alps, from sinistral transtension in Late Jurassic to dextral transpression in Early Cretaceous time.

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REFERENCES

Ampferer, O., Tektonische Nachbarschaft Karwendel-Sonnwend Gebirge, *Sitzungber. Akad. Wiss.,Wien, Math. Naturwiss. Kl. Abt. 1*, 150, 180-199, 1941.

Bernoulli, D., and H.C. Jenkyns, A Jurassic Basin: The Glasenbach Gorge, Salzburg, Austria, *Verh. Geol. Bundesanst. Austria*, 4, 504-531, 1970.

Bernoulli, D., and H.C. Jenkyns, Alpine, Mediterranean, and central Atlantic Mesozoic facies in relation to the early evolution of the Tethys, in Modern and Ancient Geosynclinal Sedimentation, *Soc. Econ. Paleontol. Mineral., Spec. Publ. 19*, 129-160, 1974.

Besse, J., and V. Courtillot, Revised and synthetic apparent polar wander Paths of the African, Eurasian, North American and Indian plates, and true polar wander since 200 Ma, *J. Geophys. Res., 96*, 4029-4050, 1991.

Blau, J., and T. Schmidt, Tektonisch kontrollierte Sedimentation im Unterlias der Lienzer Dolomiten (Österreich, Osttirol, Kärnten), *Mitt. Ges. Geol. Bergbaustud. Österr.* 34/35, 185-207, 1988.

Bosellini, A., H. Lobitzer, R. Brandner, W. Resch, and A. Castellarin, The complex basins of the Calcareous Alps and palaeomargins, *Abh. Geol. Bundesanst. Austria, 34*, 287-325, 1980.

Burchfiel, B.C., Eastern European Alpine system and the Carpathian orocline as an example of collision tectonics, *Tectonophysics, 63*, 31-61, 1980.

Channell, J.E.T., Paleomagnetism and continental collision in the Alpine belt and the formation of Late-tectonic extensional basins, in Collision Tectonics, edited by M.P. Coward and A.C. Reis, *Geol. Soc. London, Spec. Publ. 19*, 261-284, 1986.

Channell, J.E.T., B. D'Argenio, and F. Horvath, Adria, the African Promontory, in Mesozoic Mediterranean palaeogeography, *Earth Sci. Rev. 15*, 213-292, 1979.

Channell, J.E.T., R. Brandner, A. Spieler, and N.P. Smathers, Mesozoic paleogeography of the Northern Calcareous Alps - Evidence from paleomagnetism and facies analysis, *Geology, 18*, 828-831, 1990.

Channell, J.E.T., C. Doglioni, and J.S. Stoner, Jurassic and Cretaceous paleomagnetic data from the Southern Alps (Italy), *Tectonics*, in press, 1992.

Compagnoni, R., G.V. Dal Piaz, J.C. Hunziker, G. Gosso, B. Lombardo, and P.F. Williams, The Sesia-Lanzo zone, a slice of continental crust with alpine high pressure - low temperature assemblages in the western Italian Alps, *Rend. Soc. Ital. Min. Pet., 33*, 281-334, 1977.

Coward, M., and Dietrich, D., Alpine tectonics - an overview, in Alpine Tectonics, edited by M.P. Coward, D. Dietrich, and R.G. Park, *Geol. Soc. Spec. Publ. London, 45*, 1-29, 1989.

Dal Piaz, G.V., J.C. Hunziker, and G. Martinotti, La zona Sesia-Lanzo e l'evoluzione tettonico-metamorfica delle Alpi nordoccidentali interne, *Mem. Soc. Geol. Ital., 11*, 433-460, 1972.

Decker, K., P. Faupl, and A. Mueller, Synorogenic sedimentation on the Northern Calcareous Alps during the Early Cretaceous, in *Geodynamics of the Eastern Alps*, edited by H. W. Flügel and P. Faupl, pp. 126-141, Deuticke, Vienna, 1987.

Dercourt, J., et al., Geological evolution of the Tethys Belt from the Atlantic to the Pamirs since the Lias, *Tectonophysics,123*, 241-315, 1986.

Diersche, V., Der Radiolarit des Oberjura im Mittelabschnitt der Nordlichen Kalkalpen, *Geotekton. Forsch., 58*, 1-217, 1980.

Faupl, P., Pober, E. and M. Wagreich, Facies development of the Gosau Group of the eastern parts of the Northern Calcareous Alps during the Cretaceous and Paleogene, in *Geodynamics of the Eastern Alps*, edited by H.W. Flügel and P. Faupl, pp. 142-155, Deuticke, Vienna, 1987.

Fisher, R.A., Dispersion on a sphere, *Proc. R. Soc. London Ser. A, 217*, 295-305, 1953.

Flügel, H.W., P. Faupl, and H.J. Mauritsch, Implications on the Alpidic evolution of the eastern parts of the Eastern Alps, in *Geodynamics of the Eastern Alps*, edited by H.W. Flügel and P. Faupl, pp. 405-414, Deuticke, Vienna, 1987.

Frank, W., Evolution of the Austroalpine elements in the Cretaceous, in *Geodynamics of the Eastern Alps*, edited by H.W. Flügel and P. Faupl, pp. 379-406, Deuticke, Vienna, 1987.

Frank, W., G. Hoinkes, F. Purtscheller, and M. Thöni, The Austroalpine unit west of the Hohe Tauern: the Ötztal-Stubai complex as an example for the Eoalpine metamorphic evolution, in *Geodynamics of the Eastern Alps*, edited by H.W. Flügel and P. Faupl, pp. 179-214, Deuticke, Vienna, 1987.

Frisch, W., Tectonic progradation and plate tectonic evolution of the Alps, *Tectonophysics, 60*, 121-139, 1979.

Froitzheim, N., and G.P. Eberli, Extensional detachment faulting in the evolution of a Tethys passive continental margin, Eastern Alps, Switzerland, *Geol. Soc. Am. Bull., 102*, 1297-1308, 1990.

Garrison, R., and A.G. Fischer, Deep-water limestones and radiolarites of the Alpine Jurassic, in Depositional Environments in Carbonate Rocks, *Soc. Econ. Paleontol. Mineral., Spec. Publ. 14*, 20-56, 1969.

Hargraves, R.B. and A.G. Fischer, Remanent mag-

netism in Jurassic red limestones and radiolarites from the Alps, *Geophys. J. R. Astron. Soc.*, *2*, 34-41, 1959.

Heller, F., W. Lowrie, and A.M. Hirt, A review of palaeomagnetic and magnetic anisotropy results from the Alps, in Alpine Tectonics, edited by M.P. Coward, D. Dietrich, and R.G. Park, *Geol. Soc. Spec. Publ. London*, *45*, 399-420, 1989.

Hsü, K.J., Time and place in Alpine orogenesis - the Fernor Lecture, in Alpine Tectonics, edited by M.P. Coward, D. Dietrich and R.G. Park, *Geol. Soc. Spec. Publ. London*, *45*, 421-443, 1989.

Hsü, K.J., Exhumation of high-pressure metamorphic rocks, *Geology*, *19*, 107-110, 1991.

Hunziker J.C., J. Desmons, and G. Martinotti, Alpine thermal evolution in the central and the western Alps, in Alpine Tectonics, edited by M.P. Coward, D. Dietrich and R.G. Park, *Geol. Soc. Spec. Publ. London*, *45*, 353-367, 1989.

Kirschvink, J.L., The least-squares line and plane and the analysis of paleomagnetic data, *Geophys. J. R. Astron. Soc.*, *62*, 699-718, 1980.

Kralik, M., H. Krumm, and M. Schramm, Low grade and very low grade metamorphism in the Northern Calcareous Alps and in the Greywacke Zone: Illite-crystallinity data and isotopic ages, in *Geodynamics of the Eastern Alps*, edited by H.W. Flügel and P. Faupl, pp. 164-178, Deuticke, Vienna, 1987.

Krumm, H., R. Petschick, and M. Wolf, From diagenesis to anchimetamorphism, upper Austroalpine sedimentary cover in Bavaria and Tyrol, *Geodin. Acta* , *2* (1), 33-47, 1988.

Laubscher, H., Bewegung und Wärme in der Alpinen Orogenese, *Schweiz. Mineral. Petrogr. Mitt.*, *50*, 565-596, 1970.

Laubscher, H., and D. Bernoulli, History and deformation of the Alps, in *Mountain Building Processes*, edited by K.J. Hsü, pp. 169-180, Academic, San Diego, Calif., 1982.

Lein, R., Evolution of the Northern Calcareous Alps during Triassic times, in *Geodynamics of the Eastern Alps*, edited by H.W. Flügel and P. Faupl, pp. 85-102, Deuticke, Vienna, 1987.

Lemoine, M., and R. Trümpy, Pre-oceanic rifting in the Alps, *Tectonophysics*, *133*, 305-320, 1987.

Lowrie, W., Paleomagnetism and the Adriatic promontory: a reappraisal, *Tectonics*, *5*, 797-807, 1986.

Mandl, G.W., Jurassische Gleittektonik im Bereich der Hallstätter Zone zwischen Bad Ischl und Bad Aussee (Salzkammergut, Österreich), *Mitt. Ges. Geol. Bergbaustud. Österr.*, *28*, 55-76, 1982.

Mauritsch, H.J., and W. Frisch, Paleomagnetic data from the central part of the Northern Calcareous Alps, Austria. *J. Geophysics*, *44*, 623-637, 1978.

Mauritsch, H.J., and M. Becke, Paleomagnetic Investigations in the Eastern Alps and the Southern Border Zone, in *Geodynamics of the Eastern Alps*, edited by H.W. Flügel and P. Faupl, pp. 282-305, Deuticke, Vienna, 1987.

McElhinny, M.W., Statistical significance of the fold test in paleomagnetism. *Geophys. J. R. Astron. Soc.*, *8*, 338-340, 1964.

Miller, C., Petrology of the type locality eclogites from the Koralpe and Saualpe (Eastern Alps), Austria, *Schweiz. Mineral. Petrogr. Mitt.*, *70*, 287-300, 1990.

Nagel, K.H., Der Bau der Thiersee und Karwendelmulde (Tirol) interpretiert mit Hilfe statistischer Verfahren. *Geotekton. Forsch.*, *48*, 1-136, 1975.

Petschick, R., Zur Wärmegeschichte im Kalkalpin Bayerns und Nordtirols (Inkohlung und Illit-Kristallinität), *Frankfurter Geowiss. Arb.*, Ser. C, *10*, 259 pp., 1989.

Platt, J.P., J.H. Behrmann, P.C. Cunningham, J.F. Dewey, M. Helman, M. Parish, M.G. Shepley, S. Wallis, and P.J. Weston, Kinematics of the Alpine arc and the motion history of Adria, *Nature*, *337*, 158-161, 1989.

Plöchinger, B., Argumente für die intramalmische Eingleitung von Hallstätter Schollen bei Golling (Salzburg), *Verh. Geol. Bundesant. Austria*, *2*, 181-194, 1979.

Plöchinger, B., The Northern Limestone Alps in Salzburg, *Abh. Geol. Bundesanst. Austria*, *34*, 251-255, 1980.

Plöchinger, B., Erlauterungen zu Blatt 94: Hallein, Geologische Karte der Republik Osterreich, scale 1:50,000, 53pp., Geol. Bundesanst., Wien, 53 pp., 1990.

Pober, E., and P. Faupl, The chemistry of detrital chromium spinels and its implication for the geodynamic evolution of the Eastern Alps, *Geol. Rundsch.*, *77*, 641-670, 1988.

Preisinger, A., et al., The Cretaceous-Tertiary boundary in the Gosau basin, Austria. *Nature*, *332*, 794-799, 1986.

Ratschbacher, L., Strain, rotation, and translation of Austroalpine nappes, in *Geodynamics of the Eastern Alps*, edited by H.W. Flügel and P. Faupl, 237-243, Deuticke, Vienna, 1987.

Ratschbacher, L., W. Frisch, F. Neubauer, S.M. Schmid and J. Neugebauer, Extension in compressional orogenic belts: the Eastern Alps, *Geology*, *17*, 404-407, 1989.

Ratschbacher, L., W. Frisch, and H-G. Linzer, Lateral extrusion in the Eastern Alps, Part 2: structural analysis, *Tectonics*, *10*, 257-271, 1991.

Ring, U., L. Ratschbacher, W. Frisch, D. Biehler, and M. Kralik, Kinematics of the Alpine plate-margin: structural styles, strain and motion along the Penninic-Austroalpine boundary in the Swiss-Austrian Alps, *J. Geol. Soc., London*, *146*, 835-849, 1989.

Rubie, D.C., A thermal-tectonic model for high pressure metamorphism and deformation in the Sesia zone, western Alps, *J. Geology*, *92*, 21-36, 1984.

Schlager, W., and M. Schlager, Clastic sediments associated with radiolarites (Tauglboden-Schichten, Upper Jurassic, Eastern Alps), *Sedimentology*, *20*, 65-89, 1973.

Schmid, S.M., and R. Haas, Transition from near surface thrusting to intra-basement décollement, Schlinig thrust, Eastern Alps, *Tectonics* , *8*, 697-718, 1989.

Soffel, H., The palaeomagnetism of the Permian effusives near St. Anton, Vorarlberg (Austria) and the anticlockwise rotation of the Northern Calcareous Alps through 60°, *Neues Jahrb. Geol. Paläontol. Monatsh.*, *6*, 375-384, 1975.

Soffel, H., Paleomagnetism of Permo-Triassic Red Sandstones From the Northern Calcareous Alps, *J. Geophys.*, *45*, 447-450, 1979.

Spieler, A., Beckenentwicklung im Unter- und Mitteljura nordwestlich des Achensees (Tirol) und ihre Beziehung zur "Achentaler Schubmasse", Ph.D. thesis, 220 pp., Univ. Innsbruck, Innsbruck, Austria, 1992.

Srivastava, S.P., and C.R. Tapscott, Plate kinematics of the North Atlantic, in *The Geology of North America*, vol. M, *The Western North Atlantic Region*, edited by P.R. Vogt and B.E. Tucholke, pp. 379-404, Geological Society of America, Boulder, Colo., 1986.

Thöni, M., Zur Westbewegung der Oetztaler Masse. Räumliche und zeitliche Fragen an der Schlinigüberschiebung, *Mitt. Ges. Geol. Bergbaustud. Österr.*, *26*, 247-275, 1980.

Thöni, M., Eklogite in alpidisch amphibolitfaziell geprägten Serien des Ostalpins - Alter und Genese, TSK III, *Tektonik, Strukturgeologie, Kristallingeologie im Deutsch sprachigen Raum*, Kurzassungen, Graz, 1990.

Tollmann, A., Plattentektonische Fragen in den Ostalpen und der plattentektonische Mechanismus des mediterranen Orogens, *Mitt. Österr. Geol. Ges.*, *69*, 291-351, 1978.

Tollmann, A., Oberjurassiche Gleittektonik als Hauptformungsprozess der Hallstatter Region und neue Daten zur Gesamttektonik der Nörlichen Kalkalpen in den Ostalpen, *Mitt. Österr. geol. Ges.*, *74/75*, 167-195, 1981.

Tollmann, A., Geodynamic concepts of the evolution of the Eastern Alps, in *Geodynamics of the Eastern Alps*, edited by H.W. Flügel and P. Faupl, pp. 361-378, Deuticke, Vienna, 1987a.

Tollmann, A., Late Jurassic/Neocomian gravitational tectonics in the Northern Calcareous Alps in Austria, in *Geodynamics of the Eastern Alps*, edited by H.W. Flügel and P. Faupl, pp.112-125, Deuticke, Vienna, 1987b.

Trümpy, R., A possible Jurassic-Cretaceous transform system in the Alps and the Carpathians, in Processes in Continental Lithospheric Deformation, edited by S.P. Clark, B.C. Burchfiel, and J. Suppe, *Geol. Soc. Am., Spec. Pap.*, *218*, 93-109, 1988.

VandenBerg, J., and J.D.A. Zijderveld, Paleomagnetism in the western Mediterranean area, in *Alpine Mediterranean Geodynamics*, edited by H. Berckhemer and K. Hsü, Amer. Geophys. Union, *Geodyn. Ser.*, 7, 83-112, 1982.

Vecsei, A., W. Frisch, M. Pirzer, and A. Wetzel, Origin and tectonic significance of radiolarian chert in the Austroalpine rifted continental margin, in *Siliceous Deposits of the Tethys and Pacific regions*, edited by J.R. Hein and J. Obradovic, pp. 65-80, Springer-Verlag, New York, 1989.

Wächter, J., Jurassische Massflow-und Internbreccien und ihr sedimentär-tektonisches Umfeld im mittleren Abschnitt der Nördlichen Kalkalpen, *Bochumer Geol. Geotech. Arb.*, *27*, 1-239, 1987.

Watson, G.S., A test for the randomness of directions, *Mon. Not. R. Astron. Soc.*, geophysics supp., *7*, 160-161, 1956.

Weissert, H.J., and D. Bernoulli, A transform margin in the Mesozoic Tethys: evidence from the Swiss Alps, *Geol. Rundsch.*, *74*, 665-679, 1985.

Wernicke, B., Uniform-sense normal simple shear of the continental lithosphere, *Can. J. Earth Sci.*, *22*, 108-125, 1985.

Williams, P.F., and R. Compagnoni, Deformation and metamorphism of the Bard area of the Sesia Lanzo zone, western Alps, during subduction and uplift, *J. Metamorph. Geol.*, *1*, 117-140, 1983.

Winkler, W., Mid- to Early Cretaceous flysch and melange formations in the western part of the Eastern Alps. Paleotectonic implications, *Jahrb. Geol. Bundesanst., Austria*, *131*, 341-389, 1988.

R. Brandner and A. Spieler, Institut für Geologie und Paläontologie, Universität Innsbruck, 6020 Innsbruck, Austria.

J.E.T. Channell and J.S. Stoner, Department of Geology, University of Florida, Gainesville, FL 32611-2036.

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